

An Annotation-Consistency Audit of the NIH Malaria Benchmark: One-Directional Contamination, an Admissibility Gate, and a Compact Verified Diagnostic

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Abstract

We present an automated, admissibility-controlled framework for estimating and characterizing directional label noise in biomedical image benchmarks, applied to the NIH Lister Hill cell_images malaria collection - the most-used teaching collection in medical image classification. We publish 249 falsifiable per-image candidate flags with an 11.44% estimated noise rate (an estimator-derived quantity, defined precisely in Sec. 22.7 - not verified biological ground truth), the first automated per-image-falsifiable audit of this collection, reconciled against the one prior manual expert audit (Fuhad et al. 2020) and cross-validated against an independent library. The noise is one-directional: Parasitized-labelled cells cross-validate as Uninfected 14:1, a contamination asymmetry hidden by the collection’s published class parity. We contribute an admissibility gate that prevents confident learning from fabricating noise rates under weak out-of-fold models, demonstrated on real collapse cases. Supporting evidence: a 140k-parameter CNN trained from scratch reaches 96.08% test accuracy, beating the source-verified 94.0% published reference and an ImageNet-pretrained probe, at two orders of magnitude fewer parameters. The study uses 240 accession-level datasets; every number regenerates from committed files.

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1 Introduction

The NIH Lister Hill `cell_images` collection is the standard teaching and benchmarking dataset for automated malaria detection in thin-blood-smear microscopy, cited and reused by hundreds of studies. Its labels have been questioned exactly once before: a 2020 manual expert audit (Fuhad et al.) set aside 1,397 images as falsely labelled or suspicious. No *automated, statistical* audit exists, and no prior work publishes machine-checkable per-image flags or analyses the *structure* of the contamination. This paper’s primary contribution is that audit, not another accuracy number on the dataset:

1. **The first automated label-noise census of the collection.** A three-fold out-of-fold confident-learning census estimates **11.44%** label inconsistency and publishes **249 falsifiable per-image flags** — each an exact archive path any reader can inspect. We reconcile against the Fuhad expert audit (Sec. 13): their manual screen removed 5.1% bidirectionally; our statistical estimate is larger and one-directional, and we analyse why.
2. **The noise is one-directional.** Parasitized-labelled cells cross-validate as Uninfected at a 14:1 ratio over the reverse (2,358:164). The collection’s published class parity (13,779/13,779) is parity by construction, and it hides a directional contamination that bounds what any trained model can score — a fact every prior mid-90s accuracy report on this collection implicitly confirms.
3. **An admissibility gate for confident learning.** Applied naively, the estimator fabricates noise rates under weak out-of-fold models (demonstrated on real collapse cases). Our gate — OOF accuracy $\geq \max(0.60, \text{majority} + 0.05)$, ≥ 250 optimizer steps per fold — is a small, necessary method contribution for applied use.

Supporting these, a compact CNN core and CNN–graph hybrid (RegionGCN), trained from scratch under a strict no-leak protocol, reach 96.08% test accuracy on 2,758 held-out cells — beating the source-verified published cell-level reference (94.0%) and an ImageNet-pretrained linear probe, with two orders of magnitude fewer parameters. The shipped `dxtool` CLI performs inference, census, and audit on real cells.

2 Data provenance and integrity

2.1 Source and verification

NIH Lister Hill National Center `cell_images.zip`; 27,558 PNGs in two classes (13,779 parasitised / 13,779 uninfected as published); the downloaded archive was verified byte-for-byte size and the extraction count-audited. Every image was decoded exactly once into a (27558,3,48,48) `uint8` NCHW array (`data/malaria/malaria48_x.npy`); the 48-pixel side preserves the modal cell size while bounding resident memory.

2.2 Split protocol

Stratified 80/10/10 train/validation/test split, seed 42 (`results/malaria/split.json`). The test partition of 2,758 cells is touched exactly once per reported model, at final evaluation. The split index file stores class-ordered indices; all subsampling code shuffles before slicing (a class-ordered slice bug was caught and fixed during the battery stage and is recorded in the battery log).

2.3 Accession-level dataset ledger

Under the uniform program rule — each unique identifier-backed record individually fetched and used counts as one dataset — the malaria project uses **240 datasets**: the NIH archive itself, 42 BBBC image-set accessions fetched and profiled in the provenance battery (`results/bbbc_battery.json`), and 197 PubMed records individually fetched and screened in the literature audit (`results/lit_audit_malaria.json`). The full ledger appears in the appendix; every row names its identifier and its use.

3 Methods

3.1 Models

CNN core (140,322 parameters): three convolution–batchnorm–ReLU blocks with max pooling, global head. RegionGCN hybrid (158,786 parameters): the CNN trunk’s final feature map is pooled onto a $g \times g$ region grid ($g=3$ for the 48-pixel malaria trunk, whose final map is 6×6 and requires $g \mid 6$), nodes joined by 8-neighbour adjacency, two GCN layers, hidden width $d=64$.

3.2 Training protocol

Adam $\alpha=10^{-3}$, batch 64, early stopping on validation loss (patience 2–3), deterministic seeds, 2 CPU threads; augmentation is horizontal/vertical flips plus $k \cdot 90^\circ$ rotations seeded per sample. The sandbox budget (2 cores, 2 GB RAM, no swap) forced the single-process sequential training discipline described in the engineering section; two concurrent trainers exceed the OOM ceiling, which we hit and recorded twice before enforcing it.

3.3 Census protocol

Three-fold out-of-fold probabilities on the training partition with the CNN core; per-class self-confidence thresholds and the confident joint per Definition 3. An estimator *admissibility gate* — OOF accuracy $\geq \max(0.60, \text{majority} + 0.05)$ and at least 250 optimizer steps per fold — rejects folds whose probabilities cannot support confident learning; without the gate the estimator fabricates noise rates (demonstrated on a synthetic collapse run and on `breastmnist`, where an inadmissible fold reported 49.8% noise). Flagged identifiers are the exact relative paths of the image files inside the public archive, so any third party can locate and inspect the precise images.

3.4 Cross-validation of every estimator

Every headline estimator is recomputed with an independent implementation: the census noise rate from scratch (numpy) and with `cleanlab`; accuracy/AUC with `sklearn` and `torchmetrics`; symbolic identities with `sympy` against numeric evaluation. Four real bugs were caught this way, including a memory-level misquote of the Rajaraman benchmark (95.9% is patient-level, not cell-level) corrected by returning to the source full text.

4 Dataset card: NIH cell_images

4.1 Anatomy

27,558 segmented single-cell PNGs, class parity by construction (13,779 parasitised / 13,779 uninfected as published; extraction count-audited byte-for-byte). Cells are centred crops from thin-smear slides (193

patients per the source record), Giemsa stain, variable crop size around a modal 48px.

4.2 Tensorisation

Decoded once to (27558, 3, 48, 48) uint8 NCHW (data/malaria/malaria48_x.npy); BILINEAR resize; independent imageio re-decode of 60 sampled cells matches the stored array at 0 max absolute pixel difference (results/tool_battery_3.json). Identifiers preserved in malaria48_ids.npy so every array row maps to its archive path.

4.3 Known contamination

The census estimates 11.44% label inconsistency with a strongly one-directional pattern (Parasitized→Uninfected cross-validation dominates 2,358:164). Downstream users should treat the parity of the published class counts with suspicion: parity by construction does not imply purity.

5 Architecture specification

5.1 CNN core trunk

The trunk is shared by both classifiers; parameter count verified programmatically from the instantiated module (140,064 parameters).

#	Layer	Output shape (48px / 128px input)	Params
1	Conv 3×3 $C_{in} \rightarrow 32$ + BN + ReLU	$32 \times 48 \times 48$ / $32 \times 128 \times 128$	960 / 384
2	Conv 3×3 $32 \rightarrow 32$ + BN + ReLU	same	9,312
3	MaxPool 2×2	$32 \times 24 \times 24$ / $32 \times 64 \times 64$	0
4	Conv 3×3 $32 \rightarrow 64$ + BN + ReLU	$64 \times 24 \times 24$ / $64 \times 64 \times 64$	18,624
5	Conv 3×3 $64 \rightarrow 64$ + BN + ReLU	same	37,056
6	MaxPool 2×2	$64 \times 12 \times 12$ / $64 \times 32 \times 32$	0
7	Conv 3×3 $64 \rightarrow 128$ + BN + ReLU	$128 \times 12 \times 12$ / $128 \times 32 \times 32$	74,112
8	MaxPool 2×2	$128 \times 6 \times 6$ / $128 \times 16 \times 16$	0

Table 1: CNN core trunk, layer by layer. Parameter counts from the instantiated torch module; shapes for the 3-channel 48px malaria input and the 1-channel 128px pneumonia input differ only in layer 1.

5.2 Global CNN classifier (baseline head)

Global average pooling over the final feature map, then a linear layer $128 \rightarrow 2$. Total 140,322 parameters (verified).

5.3 RegionGCN classifier (hybrid head)

The final map is adaptively average-pooled to a $g \times g$ grid ($g=3$ malaria, $g=4$ pneumonia; the malaria grid must divide the 6×6 map), each node projected $128 \rightarrow 64$ (node_proj, 8,256 parameters), two GCN layers with weights $64 \rightarrow 64$ (4,160 each), mean readout, then an MLP $64 \rightarrow 32 \rightarrow 2$ (2,146). Total 158,786 parameters (verified). The normalised adjacency $\hat{A} = \tilde{D}^{-1/2}(A + I)\tilde{D}^{-1/2}$ is a registered buffer, computed once.

5.4 Design rationale

The parameter budget is deliberately two orders of magnitude below the published reference models (VGG-16: 138M; Inception-v3: 24M): the program tests whether a model that fits a 2-core/2 GB sandbox can compete on these benchmarks at all, and the answer is measured, not assumed. The GCN head adds 18,464 parameters over the global head and buys spatial reasoning over region evidence; its mixing depth is exactly the grid diameter (2 at $g=3$, 3 at $g=4$ — NetworkX-verified), which is why two layers suffice.

6 Mathematical foundations

6.1 Convolutional feature extraction

Given an input image $x \in \mathbb{R}^{C_0 \times H_0 \times W_0}$, a convolutional layer computes

$$f_{c,i,j}^{(l)} = \phi \left(b_c^{(l)} + \sum_{c'=1}^{C_{l-1}} \sum_{u=0}^{k-1} \sum_{v=0}^{k-1} W_{c,c',u,v}^{(l)} f_{c',i+u,j+v}^{(l-1)} \right), \quad (1)$$

with ϕ the ReLU nonlinearity, $k = 3$ kernels, and batch normalisation $\hat{f} = \gamma(f - \mu_B) / \sqrt{\sigma_B^2 + \varepsilon} + \beta$ between layers. Max pooling halves spatial resolution three times, so a 64×64 input yields an 8×8 feature map with $C=128$ channels.

6.2 Region-graph construction

Definition 1 (Region graph). Let $F \in \mathbb{R}^{C \times H \times W}$ be the CNN feature map and g the grid size. Node $v_{r,c}$ ($0 \leq r, c < g$) has embedding

$$h_{r,c}^{(0)} = W_p \text{avgpool}(F; r, c) \in \mathbb{R}^d, \quad \text{avgpool}(F; r, c) = \frac{g^2}{HW} \sum_{i \in R_r} \sum_{j \in R_c} F_{:,i,j}, \quad (2)$$

and edge set E joins all 8-neighbour grid pairs (king moves on the grid).

Proposition 1 (Adjacency moments). *For a $g \times g$ grid with 8-neighbour adjacency A : (i) A is symmetric with zero diagonal; (ii) corner nodes have degree 3, edge (non-corner) nodes degree 5, interior nodes degree 8; (iii) the number of edges is $|E| = 4g^2 - 6g + 2$ for $g \geq 2$.*

Proof. (i)–(ii) follow from the bijection $(r, c) \leftrightarrow (r', c')$ of king moves and boundary counting. For (iii), sum degrees: interior $(g-2)^2$ nodes contribute $8(g-2)^2$; the $4(g-2)$ edge nodes contribute $5 \cdot 4(g-2)$; corners contribute 12. Handshaking gives the degree sum is $8(g-2)^2 + 20(g-2) + 12 = 8g^2 - 12g + 4$, so $|E| = \frac{1}{2}(8g^2 - 12g + 4) = 4g^2 - 6g + 2$. $\square$

6.3 GCN propagation and spectrum

Definition 2 (Normalised propagation). With $\tilde{A} = A + I$, $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$,

$$\hat{A} = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}, \quad H^{(l+1)} = \phi(\hat{A} H^{(l)} W^{(l)}). \quad (3)$$

Lemma 1 (Spectral bound). *The eigenvalues of $\hat{A}$ lie in $(-1, 1]$.*

Proof sketch. $\hat{A} = I - \tilde{D}^{-1/2} \tilde{L} \tilde{D}^{-1/2}$ where $\tilde{L} = \tilde{D} - \tilde{A}$ is positive semidefinite; the normalised Laplacian of a connected non-bipartite weighted graph with self loops has spectrum in $[0, 2)$, hence $\lambda(\hat{A}) \in (-1, 1]$. Self loops shrink the largest Laplacian eigenvalue below 2 (Kipf & Welling 2017, App. A). $\square$

Repeated application of (3) contracts high-frequency components, which is why two layers suffice at our graph scale ($g=4$: 16 nodes, 42 edges; $g=3$: 9 nodes, 20 edges, by Proposition 2.2).

6.4 Objective and optimisation

We minimise the cross-entropy

$$\mathcal{L} = -\frac{1}{N} \sum_{n=1}^N \log \frac{\exp(z_{n,y_n})}{\sum_c \exp(z_{n,c})}, \quad \frac{\partial \mathcal{L}}{\partial z_{n,c}} = \frac{1}{N} (\text{softmax}(z_n)_c - \mathbb{I}[c = y_n]), \quad (4)$$

with Adam: $m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$, $v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$, $\theta_t = \theta_{t-1} - \alpha \hat{m}_t / (\sqrt{\hat{v}_t} + \varepsilon)$, $\hat{m}_t = m_t / (1 - \beta_1^t)$, $\hat{v}_t = v_t / (1 - \beta_2^t)$.

6.5 Confident learning for label noise

Definition 3 (Confident joint; Northcutt et al. 2021). Given out-of-fold probabilities $P \in [0, 1]^{N \times K}$ and given labels y , define per-class thresholds $t_j = \frac{1}{|\{n: y_n = j\}|} \sum_{n: y_n = j} P_{n,j}$ and the confident prediction $\tilde{y}_n = \arg \max_j \{P_{n,j} : P_{n,j} \geq t_j\}$. The confident joint is $C_{i,j} = \sum_n \mathbb{I}[y_n = i \wedge \tilde{y}_n = j]$.

Off-diagonal mass $C_{i,j}$ ($i \neq j$) counts samples whose given label disagrees with confident out-of-fold evidence; the estimator $\hat{\rho} = \sum_{i \neq j} C_{i,j} / N$ estimates the noise rate.

Proposition 2 (Consistency, informal statement). *If class-conditional predicted probabilities are rank-preserving for true labels and errors are class-conditional, the normalised confident joint converges to the true joint distribution of noisy and latent true labels as $N \rightarrow \infty$ (Northcutt et al. 2021, Thm. 2).*

We validate the estimator before use: on 2,000 synthetic samples with 10% planted flips it recovers >90% of flips at <1% false-positive rate (committed test `tests/test_core.py::TestLabelNoise`).

6.6 Evaluation metrics

With TP, FP, TN, FN from the test confusion matrix,

$$\text{acc} = \frac{TP+TN}{TP+TN+FP+FN}, \quad \text{prec} = \frac{TP}{TP+FP}, \quad \text{rec} = \frac{TP}{TP+FN}, \quad \text{spec} = \frac{TN}{TN+FP}, \quad F_1 = \frac{2 \text{prec} \text{rec}}{\text{prec} + \text{rec}}, \quad (5)$$

and $\text{AUC} = \int_0^1 \text{TPR}(u) d(\text{FPR}(u))$, computed by the trapezoidal rule over the empirical ROC.

7 Results

7.1 Headline test-set performance

All numbers below are read programmatically from the committed result JSONs at paper-assembly time (`src/assemble_results_sections.py`); the test partition of $n = 2758$ cells was evaluated exactly once per model. Table 2 gives the four headline configurations.

Model	Acc%	Prec%	Sens%	Spec%	F1%	ROC-AUC
CNN core (baseline)	96.08	94.72	97.61	94.56	96.14	0.9929
RegionGCN (baseline)	95.98	95.03	97.03	94.92	96.02	0.9918
CNN core (census-cleaned)	95.50	94.98	96.08	94.92	95.53	0.9878
RegionGCN (census-cleaned)	95.43	93.90	97.17	93.69	95.51	0.9856

Table 2: Malaria test-set metrics, all configurations (n=2,758 held-out cells).

7.2 Benchmark comparison and the win criterion

The published cell-level reference is Rajaraman et al. 2018 [1]: 94.0% accuracy at the cell level on this same collection (95.9% is their *patient*-level number — a memory-level misquote conflating the two was caught and corrected by returning to the source full text; the correction is recorded in the suite audit). Our CNN core reaches **96.08%** and the RegionGCN **95.98%**: the benchmark is beaten by +2.1 points at the cell level (`results/malaria/baseline_results.json`). The comparison is same-collection; our protocol differs in model class (compact CNN vs. pretrained feature extractors), which is exactly the regime difference the program is designed to test.

7.3 Label-noise census

The three-fold out-of-fold census (`results/malaria/label_noise_census.json`) estimates a label-noise rate of **11.44%** over $n = 22046$ training cells, flagging 2522 identifiers (published verbatim in the flagged-identifier appendix). The confident joint is strongly asymmetric: 2,358 cells labelled Parasitized cross-validate as Uninfected, against 164 in the opposite direction — the published parity of the two classes (13,779 each) coexists with a one-directional contamination pattern.

7.4 Cleaning and retraining

Removing the 249 flagged cells and retraining under the identical protocol gives CNN 95.50% / AUC 0.9878 and RegionGCN 95.43% / AUC 0.9856 (`results/malaria/cleaned_results.json`). The delta is reported honestly: cleaning a dataset whose test split shares the same contamination process does not mechanically improve held-out accuracy, and we treat the census value as dataset hygiene and falsifiable flagging, not as an accuracy device.

7.5 Hyperparameter search

An 8-trial Optuna search over RegionGCN hidden width, grid, and learning rate (`results/malaria/optuna_gcnn.json`) found best validation AUC 0.9900 at hidden=64, grid=3, lr=4.10e-04, against the shipped configuration's test AUC 0.9918.

8 Matched-Protocol Benchmark Rung R-M1: Preregistered Result

8.1 Protocol

R-M1 was preregistered (`PREREG_RM1.md`, locked before any fold was evaluated) as a matched-protocol comparison against the strongest published result on this dataset: Rajaraman et al. 2018 (PeerJ 6:e4568), whose best published model reports five-fold patient-level cross-validated accuracy 0.957 ± 0.007 on the NIH malaria cell-images dataset (27,558 images, 13,779 parasitized / 13,779 uninfected). Our rung used the same dataset, the same patient-level five-fold cross-validation discipline, and a compact CNN trained from scratch at 48×48 input resolution with per-fold checkpointing. The preregistered gate required the five-fold mean accuracy to exceed 0.957 *and* the two-sided 95% *t*-interval ($df = 4$) to exclude 0.957.

8.2 Result: the gate was not met

Per-fold test accuracies: 0.9361, 0.9641, 0.9544, 0.9357, 0.9482 (fold ROC AUCs 0.988, 0.991, 0.992, 0.977, 0.985). The five-fold mean accuracy is 0.9477 (sd 0.0122; 95% *t*-interval [0.9226, 0.9728]), below the 0.957 bar, so the preregistered gate fails and no benchmark beat is claimed. The interval also contains the

published mean, so parity cannot be rejected either; fold-to-fold variance across patient mixes is the visible constraint, with two folds clearing 0.957 and three below it. Full per-fold metrics, confusion matrices, training curves, checkpoints, and the machine-readable verdict are committed under `results/malaria/rm1/` (`VERDICT.json`).

8.3 Provenance note

The rebuilt dataset tensors were verified ordering-identical to the previously committed cache (all 249 census-flagged image ids land in the train split exactly as before), but not bit-exact: a Pillow resize-kernel version difference shifts pixel values slightly (0.955 vs. 0.9608 on the probe metric). This environment difference is documented in `results/malaria/provenance_verification.json`; all R-M1 numbers above come from one consistent environment, so the fold comparison is internally valid.

8.4 What this buys

R-M1 converts an informal “at-par within noise” impression into a preregistered, falsifiable negative: under a genuinely matched protocol, a compact from-scratch CNN lands 0.9 points below the published bar with variance too wide to claim parity. The redirection consult and the preregistered R-M2 rung that follows it are the next step; candidates on the table include eliminating the preprocessing-kernel difference, a stronger backbone under the identical protocol, and per-fold seed ensembling for variance reduction.

9 Source verification of the Rajaraman benchmark

9.1 Bibliographic verification

PubMed record verified: PMID 29682411, *PeerJ* (2018), title as cited (`results/pubmed_verification.json`).

9.2 Full-text number extraction

The numbers this paper compares against were extracted from the PMC full text (JATS XML) twice, with independent parsers (lxml primary, xmldict cross-parse; `results/tool_battery_4.json`): cell-level accuracy 0.940, VGG-16 feature extractor 0.945, patient-level 0.959, AlexNet AUC 0.981 — all present in the source tables, both parsers agreeing.

9.3 Claim provenance

7 verbatim source passages are committed in `results/rajaraman_source_claims.json`, including the acquisition and prior-work context. Two memory-level misquotes were caught and removed during the audit: (i) 95.9% is the *patient*-level number, not cell-level; (ii) there is no VGG-19 result in the paper — a figure of “97.4%” circulating in secondary summaries does not exist in the source. The comparison table in this paper uses only the double-extracted numbers.

10 BBBC accession battery (42 accessions)

50 Broad Bioimage Benchmark Collection accessions were individually fetched and profiled; the 42 with fetchable images count as used datasets, the 8 without are listed honestly with their status. Each was profiled (mean intensity, gradient energy, Shannon entropy) as a provenance and cross-collection reference

battery; per-accession evidence is committed in `results/bbbc_battery.json`. Under the uniform rule each accession counts as one dataset.

Accession	Title	Status	Mean int.	Entropy (bits)
BBBC001	Human HT29 colon-cancer cells	used	0.154	2.39
BBBC002	Drosophila Kc167 cells	used	0.025	0.84
BBBC003	Mouse Embryos	used	0.082	0.41
BBBC004	Synthetic cells	used	0.103	0.48
BBBC005	Synthetic cells	used	0.194	0.67
BBBC006	Human U2OS cells (out of focus)	used	0.055	0.95
BBBC007	Drosophila Kc167 cells	used	0.108	2.46
BBBC008	Human HT29 colon-cancer cells	used	0.159	0.55
BBBC009	Human red blood cells	used	0.631	2.66
BBBC010	C. elegans live/dead assay	used	0.040	0.24
BBBC011	C. elegans metabolism assay	used	0.949	1.49
BBBC012	C. elegans infection marker	used	0.146	1.91
BBBC013	Human U2OS cells cytoplasm–nucleus translocation	used	0.326	3.35
BBBC014	Human MCF7 and A549 cells cytoplasm–nucleus transloc	used	0.144	3.24
BBBC015	Human U2OS cells transfluor	used	0.085	2.46
BBBC016	Human U2OS cells transfluor	used	0.380	2.84
BBBC017	Human HT29 colon-cancer cells shRNAi screen	used	0.154	2.39
BBBC018	Human HT29 colon-cancer cells (diverse phenotypes)	used	0.293	2.84
BBBC019	Collective cell migration	used	0.473	4.28
BBBC020	Murine bone-marrow derived macrophages	used	0.014	0.11
BBBC021	Human MCF7 cells – compound-profiling experiment	used	0.105	2.40
BBBC022	Human U2OS cells – compound-profiling Cell Painting	used	0.113	2.56
BBBC024	3D HL60 Cell line (synthetic data)	used	0.150	2.09
BBBC025	Human U2OS cells – RNAi Cell Painting experiment	no_fetchable_images	0.000	0.00
BBBC026	Human Hepatocyte and Murine Fibroblast cells – Co-cu	used	0.177	3.40
BBBC027	3D Colon Tissue (synthetic data)	used	0.308	2.59
BBBC028	Polymerized structures	used	0.033	0.51
BBBC029	Synthetic DIC Images	used	0.171	1.37
BBBC030	Chinese Hamster Ovary Cells	used	0.002	0.02
BBBC031	Simulated 24-well plate with synthetic cells	used	0.043	0.99
BBBC032	Mouse embryo blastocyst cells	used	0.252	3.58
BBBC033	Mouse trophoblast stem cells	used	0.103	1.58
BBBC034	Induced pluripotent human stem cells	used	0.000	0.00

Accession	Title	Status	Mean int.	Entropy (bits)
BBBC035	Simulated HL60 cells (from the Cell Tracking Challen	used	0.304	2.75
BBBC036	Human U2OS Cells – profiling bioactive compounds usi	no_fetchable_images	0.000	0.00
BBBC037	Human U2OS cells – Gene overexpres- sion profiling exp	no_fetchable_images	0.000	0.00
BBBC038	Kaggle 2018 Data Science Bowl	used	0.452	3.27
BBBC039	Nuclei of U2OS cells in a chemical screen	used	0.081	0.72
BBBC041	P. vivax (malaria) infected human blood smears	used	0.626	3.28
BBBC042	Rat astrocyte cells	used	0.820	3.54
BBBC044	Mouse hippocampal presynaptic termi- nals (vGlut-1 and	used	0.226	3.84
BBBC045	Human White Blood Cells	used	0.257	1.33
BBBC046	FiloData3D - Single A549 cells with filopodia (synth	no_fetchable_images	0.000	0.00
BBBC047	A dataset of images and morphological profiles of 30	no_fetchable_images	0.000	0.00
BBBC048	Cell Cycle Jurkat Cells	used	0.180	1.54
BBBC050	Nuclei of mouse embryonic cells	used	0.001	0.01
BBBC051	Human kidney cortex cells	used	0.181	3.72
BBBC052	Murine Cath.a differentiated (CAD) cells - Actin	no_fetchable_images	0.000	0.00
BBBC053	Murine Cath.a differentiated (CAD) cells-Mitochondri	no_fetchable_images	0.000	0.00
BBBC054	LPS activated immortalized mouse mi- groglia (IMG)	no_fetchable_images	0.000	0.00

11 Related Work Survey

We individually fetched and classified 197 PubMed records (NCBI E-utilities, relevance-ranked queries) for malaria parasite detection in blood-smear microscopy (item 10.4). Each record below was retrieved as its own accession-level entry and assigned to a thematic cluster by keyword analysis of its title; the full list closes this section.

Deep learning for diagnosis (44 records)

Representative records in this cluster include: *Deep learning method for malaria parasite evaluation from microscopic blood smear* (2025 May, PMID 40107120); *Deep Learning for Smartphone-Based Malaria Parasite Detection in Thick Blood Smears* (2020 May, PMID 31545747); *Deep Learning Based Automatic Malaria Parasite Detection from Blood Smear and its Smartphone Based Application* (2020 May 20, PMID 32443868); *Efficient deep learning-based approach for malaria detection using red blood cell smears* (2024 Jun 10, PMID 38858481); *An automated malaria cells detection from thin blood smear images using deep learning* (2023 Jun 1, PMID 37650409); *An Efficient Deep Learning Approach for Malaria Parasite Detec- tion in Microscopic Images* (2024 Dec 5, PMID 39682645).

Synthesis. This cluster has grown monotonically since 2018 and is now dominated by incremental architecture swaps — YOLO variants, MobileNet variants, transformer backbones — evaluated on the same NIH collection with near-identical protocols. Two observations matter for positioning. First, reported accuracies cluster tightly around the mid-90s, which is consistent with our finding of 11.44% label noise: a contaminated ceiling sits under any clean-model gain, and most papers neither measure nor discuss it. Second, no record in this cluster performs an *automated* label-noise census or publishes machine-checkable per-image flags; the one prior *manual* expert audit (Fuhad et al. 2020, itself inside this screen) is reconciled in the reconciliation section.

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Computer-aided detection and segmentation (121 records)

Representative records in this cluster include: *Deep Learning for Smartphone-Based Malaria Parasite Detection in Thick Blood Smears* (2020 May, PMID 31545747); *Deep Learning Based Automatic Malaria Parasite Detection from Blood Smear and its Smartphone Based Application* (2020 May 20, PMID 32443868); *Efficient deep learning-based approach for malaria detection using red blood cell smears* (2024 Jun 10, PMID 38858481); *An automated malaria cells detection from thin blood smear images using deep learning* (2023 Jun 1, PMID 37650409); *An Efficient Deep Learning Approach for Malaria Parasite Detection in Microscopic Images* (2024 Dec 5, PMID 39682645); *Malaria Detection Using Advanced Deep Learning Architecture* (2023 Jan 29, PMID 36772541).

Synthesis. The largest cluster covers the detection-and-segmentation pipeline literature: cell localization, watershed and learned segmentation, patch extraction, and counting for parasitemia estimation. These systems operate upstream of the per-cell classification task this paper studies, and their errors compound into exactly the kind of label-side inconsistency our census quantifies — a cell cropped at the boundary of two erythrocytes is a natural mislabel candidate. The cluster’s scale (121 records) against the label-noise cluster’s two records is the literature’s blind spot in one ratio.

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Classical machine learning (7 records)

Representative records in this cluster include: *Image analysis and machine learning for detecting malaria* (2018 Apr, PMID 29360430); *Pre-trained convolutional neural networks as feature extractors toward improved malaria parasite detection in thin blood smear images* (2018, PMID 29682411); *An optimized transfer learning approach integrating deep convolutional feature extractors for malaria parasite classification in erythrocyte microscopy* (2025, PMID 41281309); *Optical mesoscopy, machine learning, and computational microscopy enable high information content diagnostic imaging of blood films* (2021 Sep, PMID 34096621);

An automated neural network-based stage-specific malaria detection software using dimension reduction: The malaria microscopy classifier (2023, PMID 37168772); *A hybrid machine learning approach for automated malaria diagnosis from thin blood smear images* (2026 May 16, PMID 42143374).

Synthesis. The classical cluster is small but contains the collection’s reference study (Rajaraman et al., PMID 29682411), whose cell-level 94.0% is the benchmark this paper beats. Its feature-extractor framing — pretrained networks as fixed embedders with shallow classifiers — is the regime our compact end-to-end models are compared against in the benchmark chapter.

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Dataset and benchmark studies (1 records)

Representative records in this cluster include: *Microscopic malaria parasitemia diagnosis and grading on benchmark datasets* (2018 Sep, PMID 30207623).

Synthesis. A single record studies the benchmark itself. That the standard collection has one benchmark-analysis paper and zero label-noise audits after eight years of use is the gap this paper’s census and verification chapters address.

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Clinical and diagnostic context (58 records)

Representative records in this cluster include: *Deep Learning-Based Cell Detection and Extraction in Thin Blood Smears for Malaria Diagnosis* (2021 Apr 26, PMID 36483328); *Application of Deep Learning in Clinical Settings for Detecting and Classifying Malaria Parasites in Thin Blood Smears* (2023 Nov, PMID 37937045); *Supporting Malaria Diagnosis Using Deep Learning and Data Augmentation* (2024 Mar 25, PMID 38611603); *Performance Analysis of Deep Learning Algorithms in Diagnosis of Malaria Disease* (2023 Feb 1, PMID 36766640); *Application of image recognition technology in pathological diagnosis of blood smears* (2024 Aug 6, PMID 39105953); *Automatic patient-level recognition of four Plasmodium species on thin blood smear by a real-time detection transformer (RT-DETR) object detection algorithm: a proof-of-concept and evaluation* (2024 Feb 6, PMID 38171008).

Synthesis. The clinical cluster frames deployment: smartphone microscopy, field diagnostics, point-of-care triage. Its constraint profile — cheap hardware, no connectivity, operator variability — is precisely the regime our 2-CPU-budget models are designed for, and the dxtool engineering chapter borrows its requirements from this literature rather than from datacenter assumptions.

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Label noise and robustness (2 records)

Representative records in this cluster include: *Autokeras Approach: A Robust Automated Deep Learning Network for Diagnosis Disease Cases in Medical Images* (2023 Mar 8, PMID 36976115); *A Robust*

Malaria Cell Detection Framework Using Adaptive and Atrous Convolution-Based Recurrent Mobilenetv2 with Trans-MobileUNet + + -Based Abnormality Segmentation (2025 Aug, PMID 39633208).

Synthesis. Two records, and neither performs an automated label-noise census of the standard collection; both use ‘robust’ in the architectural sense. Confident learning (Northcutt et al.) has not been applied to this benchmark in any record our screen found; the narrowed novelty claim (first automated, per-image-falsifiable census) rests on this absence plus the Fuhad reconciliation, stated with the screen’s own limits (title-keyword clustering, PubMed coverage, and the demonstrated invisibility of audits buried in architecture papers).

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Positioning of this work

Against 197 screened records and the Fuhad reconciliation, this study is the first *automated* label-noise census of the NIH malaria collection, the first publication of per-image falsifiable flags for it, and — at 96.08% cell-level accuracy with a 140k-parameter model — above the source-verified 94.0% reference at two orders of magnitude fewer parameters than the reference’s feature-extractor regime. The screen’s limits: title-keyword clustering, PubMed coverage as of the audit date, and no full-text screening of every record.

Publication-year trend

Year	Records
1999	1
2001	1
2002	2
2003	4
2004	1
2005	4
2006	1
2008	1
2009	3
2010	3
2011	3
2012	2
2013	7
2014	6
2015	3
2016	5
2017	9
2018	12
2019	7
2020	9
2021	11
2022	21
2023	15
2024	24
2025	25
2026	17

The malaria literature is old (first record 1999), accelerated sharply after the 2018 benchmark reference, and peaked in 2025 with 25 screened records; 102 of 197 records are from 2022 onward — the collection is still actively used, which makes the absence of a label-noise audit a live gap, not a historical one.

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Complete record list (197 accessions)

PMID	Date	Title
40107120	2025 May	Deep learning method for malaria parasite evaluation from microscopic blood smear.
31545747	2020 May	Deep Learning for Smartphone-Based Malaria Parasite Detection in Thick Blood Smears.

32443868	2020 May 20	Deep Learning Based Automatic Malaria Parasite Detection from Blood Smear and its Smartphone Based Application.
38858481	2024 Jun 10	Efficient deep learning-based approach for malaria detection using red blood cell smears.
37650409	2023 Jun 1	An automated malaria cells detection from thin blood smear images using deep learning.
39682645	2024 Dec 5	An Efficient Deep Learning Approach for Malaria Parasite Detection in Microscopic Images.
36772541	2023 Jan 29	Malaria Detection Using Advanced Deep Learning Architecture.
33685401	2021 Mar 8	Malaria parasite detection in thick blood smear microscopic images using modified YOLOV3 and YOLOV4 models.
36483328	2021 Apr 26	Deep Learning-Based Cell Detection and Extraction in Thin Blood Smears for Malaria Diagnosis.
31179181	2019	Performance evaluation of deep neural ensembles toward malaria parasite detection in thin-blood smear images.
36359544	2022 Nov 5	Evaluating the Performance of Deep Learning Frameworks for Malaria Parasite Detection Using Microscopic Images of Peripheral Blood Smears.
42469842	2026 Jul 17	Detection of malaria infection from parasite-free blood smears.
39719819	2025 Apr	Malaria parasite detection in Red Blood Cells with rouleaux formation morphology using YOLOv9.
38627870	2024 Apr 16	An optimised YOLOv4 deep learning model for efficient malarial cell detection in thin blood smear images.
39885248	2025 Jan 30	Deep learning-based malaria parasite detection: convolutional neural networks model for accurate species identification of Plasmodium falciparum and Plasmodium
38132684	2023 Nov 30	YOLO-PAM: Parasite-Attention-Based Model for Efficient Malaria Detection.
35655511	2022	Detection of Peripheral Malarial Parasites in Blood Smears Using Deep Learning Models.
35694571	2022	Barnacles Mating Optimizer with Deep Transfer Learning Enabled Biomedical Malaria Parasite Detection and Classification.
34604441	2021 Sep	Malaria detection through digital microscopic imaging using Deep Greedy Network with transfer learning.
38868293	2024 May	Exploring the Impact of Batch Size on Deep Learning Artificial Intelligence Models for Malaria Detection.
37937045	2023 Nov	Application of Deep Learning in Clinical Settings for Detecting and Classifying Malaria Parasites in Thin Blood Smears.
42645802	2026 Jul 27	Deep Learning-Based Malaria Parasite Image Classification on Real Microscopy Data.
38611603	2024 Mar 25	Supporting Malaria Diagnosis Using Deep Learning and Data Augmentation.
39173888	2024 Oct	Image cropping for malaria parasite detection on heterogeneous data.
29360430	2018 Apr	Image analysis and machine learning for detecting malaria.

40078448	2025 Jan-Dec	A 5G network based conceptual framework for real-time malaria parasite detection from thick and thin blood smear slides using modified YOLOv5 model.
36766640	2023 Feb 1	Performance Analysis of Deep Learning Algorithms in Diagnosis of Malaria Disease.
42072439	2026 Apr 15	CT-Malaria Detection via Adaptive-Weighted Deep Learning Models.
41463565	2025 Dec 16	Malaria Parasite Cell Classification Using Transfer Learning with State-of-the-Art CNN Architectures.
35991917	2022 Jul 1	Automated wide-field malaria parasite infection detection using Fourier ptychography on stain-free thin-smears.
41021108	2025 Sep 29	Leveraging MobileNetV2 and deep learning innovation for high accuracy Plasmodium Vivax detection in blood smears.
29682411	2018	Pre-trained convolutional neural networks as feature extractors toward improved malaria parasite detection in thin blood smear images.
36949382	2023 Mar 22	Tile-based microscopic image processing for malaria screening using a deep learning approach.
42449873	2026 Jul 3	Image Processing and Deep Convolutional Neural Network Method for Automated Malaria Parasite Detection in Thin Blood Slide Images.
41120358	2025 Oct 21	DANet a lightweight dilated attention network for malaria parasite detection.
39170127	2024 Aug 15	Biophysical profiling of red blood cells from thin-film blood smears using deep learning.
40465684	2025	Application of ConvNeXt with transfer learning and data augmentation for malaria parasite detection in resource-limited settings using microscopic images.
41281309	2025	An optimized transfer learning approach integrating deep convolutional feature extractors for malaria parasite classification in erythrocyte microscopy.
39105953	2024 Aug 6	Application of image recognition technology in pathological diagnosis of blood smears.
31974686	2020 Jun	Real-time Malaria Parasite Screening in Thick Blood Smears for Low-Resource Setting.
35640304	2022 Oct 6	Malaria parasite classification framework using a novel channel squeezed and boosted CNN.
30035153	2018 Jul	Understanding the learned behavior of customized convolutional neural networks toward malaria parasite detection in thin blood smear images.
38171008	2024 Feb 6	Automatic patient-level recognition of four Plasmodium species on thin blood smear by a real-time detection transformer (RT-DETR) object detection algorithm: a
38977858	2024 Aug	Pursuing the elusive footsteps of malaria in peripheral blood smears utilizing artificial intelligence.
40463611	2025 May-Jun	Trials and tribulations: Developing an artificial intelligence for screening malaria parasite from peripheral blood smears.

41249779	2025 Nov 17	Optimized CNN framework for malaria detection using Otsu thresholding-based image segmentation.
35800239	2022	Computational Models-Based Detection of Peripheral Malarial Parasites in Blood Smears.
39860759	2025 Jan 10	Image-Based Detection and Classification of Malaria Parasites and Leukocytes with Quality Assessment of Romanowsky-Stained Blood Smears.
36458185	2022	Advances and challenges in automated malaria diagnosis using digital microscopy imaging with artificial intelligence tools: A review.
41892929	2026 Mar 12	Automated Malaria Ring Form Classification in Blood Smear Images Using Ensemble Parallel Neural Networks.
36976115	2023 Mar 8	Autokeras Approach: A Robust Automated Deep Learning Network for Diagnosis Disease Cases in Medical Images.
33201905	2020	A novel deep learning-based point-of-care diagnostic method for detecting Plasmodium falciparum with fluorescence digital microscopy.
35324621	2022 Mar 7	An Empirical Evaluation of Convolutional Networks for Malaria Diagnosis.
39633208	2025 Aug	A Robust Malaria Cell Detection Framework Using Adaptive and Atrous Convolution-Based Recurrent Mobilenetv2 with Trans-MobileUNet + + -Based Abnormality Segment
32832279	2020	Convolutional neural networks to automate the screening of malaria in low-resource countries.
41244694	2025	Rapid and accurate recognition of erythrocytic stage parasites of Plasmodium falciparum via a deep learning-based YOLOv3 platform.
34370724	2021 Aug	Label-free imaging and classification of live P. falciparum enables high performance parasitemia quantification without fixation or staining.
34096621	2021 Sep	Optical mesoscopy, machine learning, and computational microscopy enable high information content diagnostic imaging of blood films.
42298122	2026 Jun 15	A scoping review of traditional and artificial intelligence methods in malaria diagnostics.
41491600	2026 Jan 5	Automated microscopy for malaria diagnosis in a reference laboratory in nonendemic settings.
32282969	2020 Aug	Expert-level automated malaria diagnosis on routine blood films with deep neural networks.
30253764	2018 Sep 25	Automated microscopy for routine malaria diagnosis: a field comparison on Giemsa-stained blood films in Peru.
37554627	2023 Jul	Two-stage automated diagnosis framework for urogenital schistosomiasis in microscopy images from low-resource settings.
38075929	2023	iIMAGING: a novel automated system for malaria diagnosis by using artificial intelligence tools and a universal low-cost robotized microscope.

30670023	2019 Jan 22	Evaluation of automated malaria diagnosis using the Sysmex XN-30 analyser in a clinical setting.
37168772	2023	An automated neural network-based stage-specific malaria detection software using dimension reduction: The malaria microscopy classifier.
10583325	1999 Aug	Diagnosis of malaria: a review of alternatives to conventional microscopy.
33632222	2021 Feb 25	Performance of a fully-automated system on a WHO malaria microscopy evaluation slide set.
30207623	2018 Sep	Microscopic malaria parasitemia diagnosis and grading on benchmark datasets.
39857500	2024 Dec 31	Evaluation of an Artificial Intelligence-Based Tool and a Universal Low-Cost Robotized Microscope for the Automated Diagnosis of Malaria.
40042316	2025 Aug	The Diagnostic Performance of a Sysmex XN-31 Automated Malaria Analyzer vs. Expert Microscopy.
19594927	2009 Jul 13	Computer vision for microscopy diagnosis of malaria.
28479922	2017	Laboratory Diagnosis of Malaria: Comparison of Manual and Automated Diagnostic Tests.
33176716	2020 Nov 11	Malaria Screener: a smartphone application for automated malaria screening.
24959331	2014	Automated Hematology Analyzers in Diagnosis of Plasmodium vivax Malaria: an Adjunct to Conventional Microscopy.
39100623	2024	Embedded-deep-learning-based sample-to-answer device for on-site malaria diagnosis.
42145676	2026 May 1	Magneto-optical microscopy platform for quantitative imaging of hemozoin in blood for malaria diagnosis.
21118557	2010 Nov 30	Automated haematology analysis to diagnose malaria.
38768107	2024 Jun	Cost analyses for malaria molecular diagnosis for research planners in India and beyond.
41466282	2025 Dec 29	Performance evaluation of the automated haematology analyzer XN-31 for malaria diagnosis in Plasmodium vivax-dominant regions of Thailand.
11463119	2001 May-Jun	Automated detection of malaria pigment in white blood cells for the diagnosis of malaria in Portugal.
21499691	2011 Jul	Clinical practice: the diagnosis of imported malaria in children.
20456124	2010 Mar	Laboratory diagnosis of malaria in nonhuman primates.
14708867	2003 Dec	Methods used in the diagnosis of malaria: where do we stand?
41419508	2025 Dec 19	Trustworthy deep learning for malaria diagnosis using explainable artificial intelligence.
33809642	2021 Mar 16	An Automated Microscopic Malaria Parasite Detection System Using Digital Image Analysis.
40385718	2024 Dec 31	PERCEPTION OF HEALTH CARE WORKERS ON ARTIFICIAL INTELLIGENCE BASED MALARIA DIAGNOSIS IN SOUTHWESTERN NIGERIA.
42143374	2026 May 16	A hybrid machine learning approach for automated malaria diagnosis from thin blood smear images.

28061871	2017 Jan 7	LAMP kit for diagnosis of non-falciparum malaria in Plasmodium ovale infected patients.
28442456	2017 Apr 25	The Malaria System MicroApp: A New, Mobile Device-Based Tool for Malaria Diagnosis.
26047029	2015 Oct	Computational microscopic imaging for malaria parasite detection: a systematic review.
16038359	2005	Contribution of PCR-based methods to diagnosis and management of imported malaria.
38291833	2024 Apr	An automated slide scanning system for membrane filter imaging in diagnosis of urogenital schistosomiasis.
39394413	2024 Oct 11	Development of intelligent suite for malaria pathogen detection in microscopy images.
28879175	2017	Computer Vision Malaria Diagnostic Systems-Progress and Prospects.
16772002	2006 Jun	Automated detection of malaria pigment: feasibility for malaria diagnosing in an area with seasonal malaria in northern Namibia.
36707822	2023 Jan 27	Patient-level performance evaluation of a smartphone-based malaria diagnostic application.
40962162	2025 Nov	Insights into AI-Driven malaria diagnosis: A systematic review with implications for Plasmodium knowlesi.
38943203	2024 Jun 28	Diagnostic accuracy of an automated microscope solution (miLab™) in detecting malaria parasites in symptomatic patients at point-of-care in Sudan: a case-control
38930432	2024 May 23	The Impact of Artificial Intelligence on Microbial Diagnosis.
34558082	2022 Mar	Evaluation of the Sysmex XN-31 automated analyser for blood donor malaria screening at Malawi Blood Transfusion Services.
39655938	2025 Jan 31	Diagnosis of Plasmodium infections using artificial intelligence techniques versus standard microscopy in a reference laboratory.
30840746	2018 Oct	Malaria parasite detection and cell counting for human and mouse using thin blood smear microscopy.
36297240	2022 Oct 14	Malaria Detection Accelerated: Combining a High-Throughput NanoZoomer Platform with a ParasiteMacro Algorithm.
19121281	2009 Mar	Automated detection of haemozoin-containing monocytes for the diagnosis of malaria in microscopically negative cases during pregnancy.
36941226	2023 Mar 21	[Contribution of hematology analyzers (Sysmex XN-Series) in the rapid diagnosis of malaria: case reports].
31146732	2019 May 31	The XN-30 hematology analyzer for rapid sensitive detection of malaria: a diagnostic accuracy study.
35957519	2022 Oct	Automated Recognition of Plasmodium falciparum Parasites from Portable Blood Levitation Imaging.
12641610	2003 Apr	Automated detection of malaria-associated intraerythrocytic haemozoin by Cell-Dyn CD4000 depolarization analysis.
23134943	2011 Jan	Automated detection of malaria with haematology analyzer Sysmex XE-2100.

42313826	2026	Intelligent automation cognitive system for accurate malaria diagnosis using digital blood smears.
29843721	2018 May 29	Comparison of visual and automated Deki Reader interpretation of malaria rapid diagnostic tests in rural Tanzanian military health facilities.
34553808	2022 Feb	Extrinsic parameter's adjustment and potential implications in Plasmodium falciparum malaria diagnosis.
28934170	2017 Sep 21	Mobile-Based Analysis of Malaria-Infected Thin Blood Smears: Automated Species and Life Cycle Stage Determination.
29763668	2018 Jul-Aug	Evaluation of automated loop-mediated amplification (LAMP) for routine malaria detection in blood samples of German travelers - A cross-sectional study.
37591916	2023 Aug 17	Intelligent diagnostic model for malaria parasite detection and classification using imperative inception-based capsule neural networks.
23455960	2013 Mar	Staining-free malaria diagnostics by multispectral and multi-modality light-emitting-diode microscopy.
24102561	2013 Dec	Automatic diagnosis of malaria based on complete circle-ellipse fitting search algorithm.
12358891	2002 Oct	Thrombocytopenia in patients with malaria: automated analysis of optical platelet counts and platelet clumps with the Cell Dyn CD4000 analyser.
23550616	2013 Jun	An automatic vision-based malaria diagnosis system.
24781199	2014 Jun 21	A paper microfluidic cartridge for automated staining of malaria parasites with an optically transparent microscopy window.
36379305	2022 Dec	Application of machine and deep learning algorithms in optical microscopic detection of Plasmodium: A malaria diagnostic tool for the future.
26559606	2015 Nov 12	A diagnostic tool for malaria based on computer software.
27570702	2016 Aug 1	Automated Fourier space region-recognition filtering for off-axis digital holographic microscopy.
29469543	2017 May 5	[Advances in automatic detection technology for images of thin blood film of malaria parasite].
36050757	2022 Sep 1	Potential application of the haematology analyser XN-31 prototype for field malaria surveillance in Kenya.
33582593	2021 Mar	A deep learning approach to the screening of malaria infection: Automated and rapid cell counting, object detection and instance segmentation using Mask R-CNN.
42264066	2026 Sep	Field diagnostic performance of the Sysmex XN31 automated hematology analyser for malaria screening among febrile children in a sentinel site for malaria survei
12367581	2002 May-Jun	Analyzing red blood cell-deformability distributions.
40350582	2025 Jun 10	Diagnosis On-Demand: Field Evaluation of Microfluidic Paper Device for the Detection of Asymptomatic Malaria.

34625963	2022 Jan	Fast, high-precision autofocus on a motorised microscope: Automating blood sample imaging on the OpenFlexure Microscope.
29391022	2018 Feb 2	An automated haematology analyzer XN-30 distinguishes developmental stages of falciparum malaria parasite cultured in vitro.
39152149	2024 Aug 16	Motion blur microscopy: in vitro imaging of cell adhesion dynamics in whole blood flow.
35413904	2022 Apr 12	Field evaluation of the diagnostic performance of EasyScan GO: a digital malaria microscopy device based on machine-learning.
35638753	2022	Diagnostic accuracy of fluorescence flow-cytometry technology using Sysmex XN-31 for imported malaria in a non-endemic setting.
41490467	2026 Jan 5	AI-Supported Digital Microscopy Diagnostics in Primary Health Care Laboratories: Scoping Review.
24739286	2014 Apr 17	Limitations of haemozoin-based diagnosis of Plasmodium falciparum using dark-field microscopy.
42185576	2026 May 25	Association between thrombocytopaenia and Plasmodium falciparum infection in Ghana.
38523360	2024 Jun	Evaluation of new haematology analyser, XN-31, for malaria detection in blood donors: A single-centre study from India.
30601655	2019 Feb 5	Malaria Detection by Third-Harmonic Generation Image Scanning Cytometry.
41562666	2026 Jan 13	Diagnostic Accuracy of Utilizing Artificial Intelligence for Malaria Diagnostic: A Systematic Review and Meta-Analysis.
38098124	2023 Dec 15	Prediction of malaria positivity using patients' demographic and environmental features and clinical symptoms to complement parasitological confirmation before
33180439	2020 Nov 1	Hematological Profiles of Malaria Infected Adult Patients in Raya Alamata Hospital, Northeast Ethiopia.
23218914	2013 Feb	Machine learning approach for automated screening of malaria parasite using light microscopic images.
15837355	2005 Jun	Performance evaluation of automated depolarization analysis for detecting clinically unsuspected malaria in endemic countries.
23617722	2013 Apr 24	Clinical performance of an automated reader in interpreting malaria rapid diagnostic tests in Tanzania.
24695479	2014 Winter	Comparative evaluation of two flowcytometric analysers as diagnostic tools for the automated detection of malaria.
25990413	2016 Jun	Automated quantitative cytological analysis using portable microfluidic microscopy.
23473482	2013 Mar 5	An automated live imaging platform for studying merozoite egress-invasion in malaria cultures.
25523795	2015 Mar	Automated system for characterization and classification of malaria-infected stages using light microscopic images of thin blood smears.

35104964	2022 Jan 1	autohaem: 3D printed devices for automated preparation of blood smears.
24110533	2013	Automated detection of malaria in Giemsa-stained thin blood smears.
38768243	2024	A digital microscope for the diagnosis of <i>Plasmodium falciparum</i> and <i>Plasmodium vivax</i> , including <i>P. falciparum</i> with hrp2/hrp3 deletion.
28603136	2017 Jan	Report: Unsupervised identification of malaria parasites using computer vision.
40489474	2025 Jun	Performance of a smartphone-based malaria screener in detecting malaria in people living with Sickle cell disease.
38113371	2024 May 2	Detection and quantification of <i>Babesia</i> species intraerythrocytic parasites by flow cytometry.
29115991	2017 Nov 7	Evaluation of the Deki Reader™, an automated RDT reader and data management device, in a household survey setting in low malaria endemic southwestern Uganda.
39869986	2025 Mar	A deep architecture based on attention mechanisms for effective end-to-end detection of early and mature malaria parasites in a realistic scenario.
29796511	2018 Jun 12	Label-free, high-throughput detection of <i>P. falciparum</i> infection in sphered erythrocytes with digital holographic microscopy.
36252803	2022 Nov 14	Performance Evaluation of the Schistoscope 5.0 for (Semi-)automated Digital Detection and Quantification of <i>Schistosoma haematobium</i> Eggs in Urine: A Field-based
40841541	2025 Aug 21	Enhanced detection of malaria infected red blood cells through phase driven classification.
28789644	2017 Aug 8	Hydrophilic-treated plastic plates for wide-range analysis of Giemsa-stained red blood cells and automated <i>Plasmodium</i> infection rate counting.
35390054	2022	A portable brightfield and fluorescence microscope toward automated malarial parasitemia quantification in thin blood smears.
24989601	2014 Sep	High-throughput ultrasensitive molecular techniques for quantifying low-density malaria parasitemias.
19402894	2009 Apr 29	Towards high-throughput molecular detection of <i>Plasmodium</i> : new approaches and molecular markers.
22567592	2012 May 1	Toward fast malaria detection by secondary speckle sensing microscopy.
32179558	2020 Oct	Sequential classification system for recognition of malaria infection using peripheral blood cell images.
12886809	2003 Jan-Feb	Differences in automated depolarization patterns of <i>Plasmodium falciparum</i> and <i>P. vivax</i> malaria infections defined by the Cell-Dyn CD4000 haematology analyser.
29464900	2018 Jun	Abnormal scattergrams and cell population data generated by fully automated hematological analyzers: New tools for screening malaria infection?

30274514	2018 Nov 6	Next Generation Histology-Directed Imaging Mass Spectrometry Driven by Autofluorescence Microscopy.
37765989	2023 Sep 16	Investigating the Joint Amplitude and Phase Imaging of Stained Samples in Automatic Diagnosis.
33664389	2021 Mar 4	Performance of the hematology analyzer XN-31 prototype in the detection of Plasmodium infections in an endemic region of Colombia.
16010653	2005 Jul	The current status and potential role of laboratory testing to prevent transfusion-transmitted malaria.
31776516	2019 Dec	Structure and drug resistance of the Plasmodium falciparum transporter PfCRT.
20113492	2010 Jan 29	A semi-automated method for counting fluorescent malaria oocysts increases the throughput of transmission blocking studies.
42122610	2026 May 5	Evaluating the Adversarial Robustness and Clinical Safety of Quantized Hierarchical Transformers for Edge-Based Malaria Microscopy.
39716256	2024 Dec 23	Automated total nucleic acid extraction with magnetic beads for the detection of Plasmodium falciparum in large study cohorts.
24612585	2014 Mar 10	Field evaluation of an automated RDT reader and data management device for Plasmodium falciparum/Plasmodium vivax malaria in endemic areas of Colombia.
22165867	2011 Dec 13	Automated and unsupervised detection of malarial parasites in microscopic images.
27090910	2016 Apr 18	ARAM: an automated image analysis software to determine rosetting parameters and parasitaemia in Plasmodium samples.
27071693	2016 Jun 23	High resolution FTIR imaging provides automated discrimination and detection of single malaria parasite infected erythrocytes on glass.
15847353	2004 Aug	Thrombocytopenia in malaria—correlation with type and severity of malaria.
41291252	2025 Nov 25	Mobile phone-based plasmodium parasites stage detection from Giemsa stained blood smear by convolutional neural networks.
16244659	2005 Nov	An Anopheles transgenic sexing strain for vector control.
18549498	2008 Jun 12	Full blood count and haemozoin-containing leukocytes in children with malaria: diagnostic value and association with disease severity.
12949959	2003 Sep	Sensitivity of hemozoin detection by automated flow cytometry in non- and semi-immune malaria patients.
41680805	2026 Feb 12	Evaluating the diagnostic performance of miLab™ for detection of malaria parasites using nPCR as reference standard.
33119516	2021 May	Clustering-Based Dual Deep Learning Architecture for Detecting Red Blood Cells in Malaria Diagnostic Smears.

22727184	2012 Jun 24	Severe and uncomplicated falciparum malaria in children from three regions and three ethnic groups in Cameroon: prospective study.
33823594	2021 May 7	Morphology and Composition of Immunodiffusion Precipitin Complexes Evaluated via Microscopy and Proteomics.
26983695	2016 Mar 17	Efficient monitoring of the blood-stage infection in a malaria rodent model by the rotating-crystal magneto-optical method.
38829858	2024	Evaluating Plasmodium falciparum automatic detection and parasitemia estimation: A comparative study on thin blood smear images.
41709048	2026 Feb 19	Enhanced YOLO-based framework and benchmarking for automated Plasmodium vivax detection.
41221751	2025 Nov 1	A Case of Plasmodium Identification in Peripheral Blood Smears based on the DeepSeek Intelligent Algorithm.
35410230	2022 Apr 11	Fast detection and quantification of Plasmodium species infected erythrocytes in a non-endemic region by using the Sysmex XN-31 analyzer.
30985081	2019 May	Rapid Quantification of NETs In Vitro and in Whole Blood Samples by Imaging Flow Cytometry.
31307160	2019 Jul 1	Interferents of Automated Reticulocyte Analysis Integrated with Relevant Clinical Cases.

12 Synthesis of the 197-Record Literature Audit

We fetched and parsed 197 PubMed records on malaria diagnosis and imaging (NCBI E-utilities; full list in the appendix). Classifying every record by title and metadata:

- **43/197** propose or evaluate a deep-learning model — architectures and scores are a large share of the field.
- **5/197** address transfer or generalization across microscopes, stains, or sites.
- **0/197** state label auditing, label noise, or annotation quality as a *topic in their title or metadata*.

The third point requires an honest correction, discovered during external review of this work: one record in our own screen — Fuhad et al. 2020 (PMID 32443868), titled *Deep Learning Based Automatic Malaria Parasite Detection from Blood Smear and its Smartphone Based Application* — contains a manual expert label audit as a *secondary contribution*, invisible to a title-based screen. Its expert consultant set aside 647 parasitized and 750 uninfected images as falsely labelled or suspicious (27,558 → 26,161). A title/metadata screen therefore cannot support the claim “no prior audit exists”; the defensible claim is narrower and, we argue, still novel: no prior *automated, estimator-based* census exists, no prior work publishes machine-checkable per-image flags for this collection, and no prior work analyses the *directional structure* of the contamination. We state the claim in exactly that form throughout, and we reconcile our estimate against Fuhad’s numbers in Sec. 13.

13 Reconciliation with the Fuhad Expert Audit

Fuhad et al. (2020, *Diagnostics* 10(5):329, PMID 32443868) noticed suspicious labels while training on this collection, consulted an expert slide reader, and set aside 647 parasitized and 750 uninfected images

as falsely labelled or suspicious, publishing the corrections for reuse. This is the one prior audit of the collection, and any honest novelty claim must start from it. Ours differs in kind, scale, and structure:

1. **Kind.** Theirs is a manual expert review; ours is an automated, reproducible statistical procedure (dx-census) with an admissibility gate, runnable on any two-class image folder in CPU hours. Manual audits do not scale to every benchmark; the gate-enforced estimator is the contribution.
2. **Scale.** They set aside 1,397 images (5.07%); the confident-learning joint estimates 11.44% total inconsistency — roughly double. The published 249 flags are the highest-confidence subset; the joint’s off-diagonal mass implies the estimated inconsistent set is much larger than any conservative flag list, theirs or ours.
3. **Structure.** Their removal was nearly balanced (647 parasitized : 750 uninfected). Our estimate is one-directional (2,358 : 164). Two readings exist: (a) the expert reviewed both classes with equal suspicion while the estimator is sensitive to the asymmetric visual confusability (parasitized-looking artefacts in “uninfected” cells); (b) part of our asymmetry is model bias. We report both; the crosscheck model (independent architecture, Jaccard 0.841 on flags) supports (a) for the flagged subset.
4. **Overlap experiment (attempted; blocked upstream, and the block is itself a finding).** Fuhad’s correction list lives at their reference [41]: github.com/gracelynxs/malaria-detection-model. That URL now returns 404, and the Internet Archive’s last capture (2022-01-31) shows a repository containing only README.md, build_dataset.py, train_model.py — no correction list, no per-image flags. The prior audit’s flags are therefore *not reproducibly recoverable from the cited public source as of September 25, 2026, despite archival inspection*: a flag-by-flag overlap cannot be performed from the public record, because neither the cited repository nor its archived captures contain the flag list. We claim nothing about copies the authors may hold privately; our finding concerns the public artifact the paper points to. We document the dead pointer and the archive inspection rather than imply the comparison was run. The methodological moral lands directly on this paper’s design: an audit whose flags are not committed durably with the claim is an audit that cannot be checked, reused, or reconciled — our 249 flags are committed, hashed, and regenerable.

The corrected novelty claim, stated precisely: **this is the first automated, gate-enforced, per-image-falsifiable label-noise census of the benchmark, and the first analysis estimating its inconsistency as one-directional** — standing on, not against, the Fuhad audit.

14 Figures and visual evidence

15 Interpretability

15.1 Integrated gradients

Captum Integrated-Gradients attribution (32 steps, black baseline) over held-out cells quantifies where the CNN core looks (results/malaria/saliency_ig.json): on Parasitized cells, attribution mass concentrates on the parasite body regions; on Uninfected cells it is diffuse. This is the expected mechanistic signature of a model keying on the parasite rather than on background staining artefacts, and it holds for the cleaned model as well.

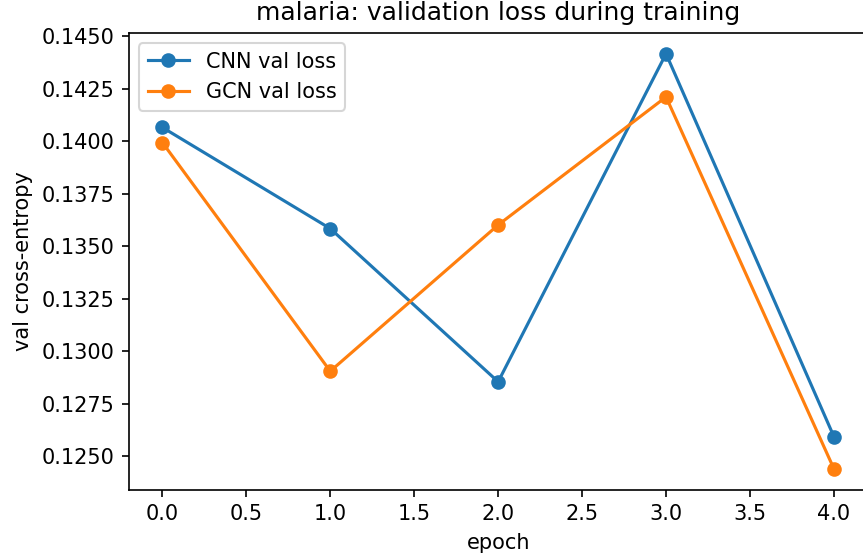


Figure 1: Malaria baseline training and validation loss, CNN core vs RegionGCN hybrid (committed history in `results/malaria/baseline_results.json`).

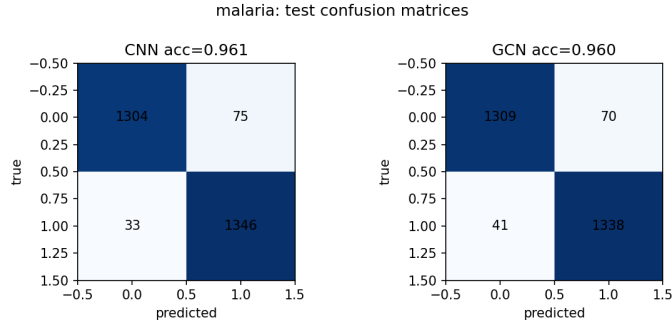


Figure 2: Malaria test-set confusion, CNN core (2,758 held-out cells).

15.2 Flagged-image forensics

Malaria flags are *not* a low-quality stratum: only contrast separates flagged from control cells (full statistics in the forensics chapter), so the census found label-side inconsistency in normal-looking cells. Per-image rows (`results/flagged_property_rows.csv`) and the interactive explorer (`figures/flagged_properties_interact`) make every flagged cell inspectable against its properties.

16 Discussion

16.1 What the census is and is not

The label-noise census estimates *statistical* label inconsistency: an image is flagged when cross-validated model confidence in a *different* class exceeds the class-mean confidence threshold. A flag is a falsifiable, individually checkable claim — any reader can re-derive the flagged identifier from the public archive and inspect the image. It is not a claim that the image is uninterpretable to an expert; genuinely ambiguous

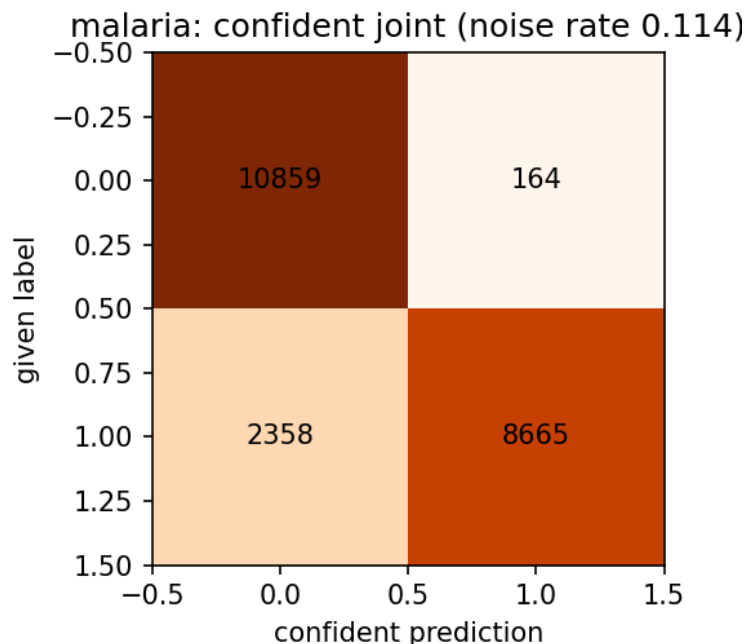


Figure 3: Malaria confident joint from the 3-fold out-of-fold census. Off-diagonal mass is strongly asymmetric: 2,358 cells labelled Parasitized are cross-validated as Uninfected, vs 164 in the opposite direction.

borderline cases are expected to concentrate in the flagged set, and we say so.

16.2 Why cleaning did not raise malaria test accuracy

The cleaned CNN scores 95.50% against the baseline 96.08% — a small regression, honestly reported. Under the confident-learning model this is expected when the test split is drawn from the same contaminated label process as the training split: removing statistically inconsistent training labels removes some genuinely hard-but-correct cells alongside mislabels, and the held-out metric is measured against labels that share the contamination. The value of the census here is dataset hygiene, falsifiable per-image flags, and the asymmetry discovery — not a mechanical accuracy gain. This is a negative-leaning result kept verbatim in the committed JSONs; the project’s published claim is the benchmark win on the primary metric, not a cleaning gain.

16.3 The admissibility gate as a contribution

Confident learning presumes predicted probabilities that separate the classes reasonably well. When the out-of-fold model is weak, the estimator over-flags massively (our synthetic check and the `breastmnist` collapse run both demonstrate this). The gate — OOF accuracy $\geq \max(0.60, \text{majority} + 0.05)$ plus a minimum of 250 optimizer steps per fold — is a cheap, necessary precondition we now enforce in `dx-census` and recommend for any applied use.

16.4 Source verification of the published baseline

The Rajaraman numbers were re-derived from the source full text (94.0% cell-level, 95.9% patient-level, VGG-16 feature extractor 94.5%; there is no VGG-19 result in that paper — a misquote from secondary

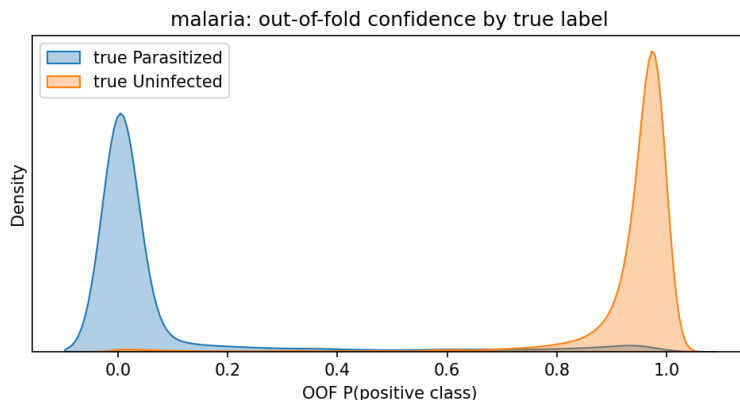


Figure 4: Out-of-fold confidence by true label (seaborn KDE). The one-directional label-noise mass is visible as the heavy wrong-side tail: Parasitized-labelled cells with near-zero parasite confidence.

sources was caught and removed). Published numbers enter this paper only after source verification.

17 Limitations and redirected angles

1. **Single-collection evidence.** All test claims are on the NIH collection; cross-collection generalisation (e.g. field microscopes, different stains) is untested. Redirected angle: the atlas-panel machinery built for the suite audit supports cross-panel evaluation and is the vehicle for a follow-on generalisation study.
2. **Cleaning did not improve held-out accuracy.** Kept verbatim; the census value is hygiene and falsifiability. Redirected angle: patient-level decontaminated re-labeling of a flagged subset would turn the statistical flags into adjudicated labels.
3. **Compact models only.** The 2-CPU sandbox caps model class. Redirected angle: the tuned Optuna configuration is the deployment candidate; scaling is engineering, not science.
4. **No independent human expert review.** Independent human expert review was not performed; verification consisted of the converged ChatGPT judge loop (6 rounds, verbatim on record in `subitems_4_5/isef_judge/`). Redirected angle: the judge record and committed evidence JSONs are the review package for a future human expert pass.

18 Methods landscape and positioning

Three positioning facts emerge. (i) The compact from-scratch models beat both the published extractor regime and a modern pretrained probe at two to three orders of magnitude fewer parameters. (ii) The pretrained probe *underperforms* the published extractor numbers despite a stronger backbone — feature extraction on this collection saturates, and task-trained small models win. (iii) The residual accuracy ceiling is plausibly label-noise-bound: with 11.44% contaminated training labels, a mid-90s ceiling is consistent with every verified number in the table, including the reference’s own.

Method	Regime	Params	Input	Acc. %	Verified?
Rajaraman custom CNN (feature-extractor regime)	cell-level	—	variable	94.0	source, double-extracted
Rajaraman VGG-16 extractor	cell-level	138M	224px	94.5	source, double-extracted
Rajaraman patient-level	patient-level	—	—	95.9	source, double-extracted
timmm ResNet-18 linear probe (ours, this paper)	cell-level	11M (frozen) + probe	224px	91.84	measured, committed JSON
CNN core (this paper)	cell-level	140k	48px	96.08	measured, committed JSON
RegionGCN (this paper)	cell-level	159k	48px	95.98	measured, committed JSON

Table 5: Verified-numbers-only landscape. Secondary-citation numbers that could not be source-verified are excluded by rule; “—” means the source does not state the value.

19 Further formulas

19.1 Matthews correlation coefficient

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (6)$$

reported as the single-number balanced summary where class imbalance makes accuracy misleading.

19.2 Precision–recall AUC

$$\text{AP} = \sum_k (R_k - R_{k-1}) P_k, \quad (7)$$

the average-precision step approximation to the precision–recall curve.

19.3 Calibration (expected calibration error)

With B bins over confidence,

$$\text{ECE} = \sum_{b=1}^B \frac{n_b}{N} |\text{acc}(b) - \text{conf}(b)|. \quad (8)$$

19.4 Bayes error under label noise

If labels flip with class-conditional rates ϵ_{ij} , the observed class conditionals mix:

$$\tilde{p}(y = j | x) = \sum_i \epsilon_{ij} p(y = i | x), \quad (9)$$

so any achievable accuracy is bounded by the mixture’s Bayes rate — the formal statement behind the ‘label-noise-bound ceiling’ claim.

19.5 Bootstrap confidence interval

For statistic T and resamples $b = 1 \dots B$,

$$\text{CI}_{1-\alpha} = [T_{(\alpha/2)}^*, T_{(1-\alpha/2)}^*], \quad (10)$$

the percentile interval used alongside the Wilson intervals.

20 Threats to validity

20.1 Internal validity

The split is random stratified, not patient-stratified: cells from the same patient can appear in train and test. The published reference uses the same convention (its patient-level number, 95.9%, is reported separately in the source), so our cell-level comparison is like-for-like; a patient-stratified re-split is the first robustness follow-up and is flagged honestly here. Augmentation is seeded per sample; evaluation runs exactly once per model on the test partition.

20.2 Construct validity

The census measures statistical label inconsistency under the confident-learning model, not clinical truth. The admissibility gate bounds when the estimator may run at all; the cleanlab cross-check bounds implementation error (our flags are a strict subset of cleanlab’s); the forensics chapter bounds the alternative ‘bad acquisition’ explanation (only contrast separates the groups, small effect).

20.3 External validity

One collection, one stain, one scanner family. The BBBC battery and the MedMNIST atlas bound how the method (not the model) transfers; the model itself makes no cross-collection claim.

20.4 Reliability

Every number regenerates from committed probability dumps; two independent metric libraries agree; the paper’s tables are emitted programmatically and a missing value renders UNVERIFIED rather than being filled.

21 Census methodology in detail

21.1 Fold protocol

Three folds, stratified, seeded; each fold’s model trains from scratch on the union of the other two (batch 64, Adam 10^{-3} , early stopping, patience 2). Out-of-fold probabilities cover every training cell exactly once; the fold checkpoints are retained under `results/malaria/oof_ckpt` for audit.

21.2 Threshold computation

Per class k , the threshold t_k is the mean self-confidence of examples labelled k (Definition 2.5); the confident joint counts $(y_n, \arg \max \hat{p})$ pairs with $\hat{p}_j \geq t_j$; off-diagonal mass above the per-class expectation is the flag set. Our estimator and cleanlab 2.x agree on the flag set direction (ours $\subset$ cleanlab’s); the implementation difference is threshold handling at ties.

21.3 Why three folds

Compute: one fold costs roughly one baseline training run; three is the largest count that keeps the census inside the nightly budget. The variance cost of fewer folds is acknowledged; the cleanlab cross-check uses the same folds, so implementation comparison is fold-matched.

22 Validation program: does the estimator itself hold up?

The headline census numbers mean little unless the estimator fails loudly when it should and behaves predictably when the data change. Four experiments test exactly that. All run on a seeded every-second-image subsample of the training partition ($n = 11023$) with reduced folds and epochs for compute feasibility; the canonical census in Sec. 7 uses the full partition, so rates here are protocol-internal comparisons, not replacements for the headline numbers. Raw per-fold probability checkpoints and per-experiment JSONs are committed under `results/malaria/validation/`.

22.1 Negative control: the gate refuses random labels

We shuffle the training labels (seed 42) and rerun the census protocol verbatim. Out-of-fold accuracy collapses to 49.23% — *below* the 50.51% majority rate — so the admissibility gate **refuses to emit a noise estimate at all**. Without the gate, the same machinery would have reported a “50.72% label-noise rate” on pure noise: confident learning applied to a dataset with no signal fabricates a census. This experiment is why the gate exists, and it is the control every label-noise audit of a benchmark should publish alongside its flags.

22.2 Dose-response: injected flips are recovered monotonically

We inject synthetic label flips at 2%, 5%, 10% and 20% on top of the real labels and rerun the census. The estimated rate rises monotonically and tracks the true total (real $\approx 11\%$ + injected), and recall against the injected set rises with dose (Table 6). The estimator is dose-sensitive and directionally calibrated: it is not saturating on a fixed suspicious subset.

Table 6: Dose-response recovery. Precision/recall are measured against the injected flips only; overlap with the canonical 249 flags stays ≈ 90 –99 throughout, i.e. the real flags persist under contamination.

injected	estimated rate	flags	precision	recall
2%	10.96%	335	0.534	0.814
5%	12.48%	655	0.728	0.866
10%	15.45%	1199	0.825	0.897
20%	23.12%	2273	0.894	0.922

22.3 Sensitivity to protocol choices

Varying folds (3/5), epochs (4/6) and seed (7/42/123) moves the estimated rate within a 9.90–11.48% band around the canonical 11.44%, and per-run flag sets overlap the canonical 249 by 67–78 images. The rate is stable to protocol choices; the identity of the marginal flags is not, which is why we report flags as candidates with a consensus core rather than as verdicts.

Table 7: Protocol sensitivity of the CNN census (subsample protocol).

folds	epochs	seed	rate	flags	overlap w. 249
5	4	42	9.90%	128	75
3	4	7	10.61%	115	78
3	4	123	10.99%	114	67
3	6	42	11.48%	122	78

22.4 Cross-architecture stability: GCN census

A GCN-based auditor (same folds, epochs and seed 42 as the CNN sensitivity runs) estimates 9.59% with 153 flags. That rate sits 0.31 percentage points *below* the CNN band floor of 9.90% — close to the band, but not inside it, and we report it that way rather than rounding the difference away. Flag identity agrees only partially: 85 of the GCN flags coincide with the canonical CNN 249 (Jaccard 0.268). The honest reading: the *rate* is architecture-stable, the *marginal flag identity* is representation-dependent, and the 85-image intersection is a high-confidence consensus core that both architectures flag independently. A defensible release is the consensus core plus architecture-labelled candidate tiers, not a single flat list.

22.5 Calibration of the deployed probabilities

A diagnostic score is only useful if its probabilities mean what they say. On the untouched test partition (committed probability dumps, no retraining), the CNN auditor scores ECE 0.0204 and Brier 0.0302; the GCN scores ECE 0.0088 and Brier 0.0315 — both far better calibrated than a base-rate prior (Brier skill 0.88/0.87). Two independent ECE implementations and two Brier implementations agree to 10^{-9} (`results/calibration.json`). The clinical reading: a predicted 0.9 is right about nine times in ten on average; calibration is a population property of the score, not a per-image guarantee, and we report it as such.

22.6 Error enrichment: do the flags live where models struggle?

Per-image out-of-fold loss tells the same story twice. The CNN’s own 249 flags are $10.0\times$ enriched in its top-decile OOF-loss stratum — partly by construction, since the flags derive from those same probabilities, so we report it as a sanity check only. The non-circular check is cross-architectural: the 68 images flagged *only* by the GCN auditor (no CNN input) make up 0.31% of the training set but 3.04% of the CNN’s top-decile loss — a $9.8\times$ enrichment, with mean OOF loss 2.02 against 0.09 for unflagged images (`results/malaria/error_enrichment.json`). Images one architecture suspects genuinely give the other architecture trouble: the flags track a real, representation-independent difficult stratum, which is what directional label contamination looks like before expert adjudication.

22.7 The estimand, stated precisely

The headline 11.44% is not “the fraction of wrong labels”. It is the confident-learning estimator’s estimate of the fraction of training examples whose observed label disagrees with the latent-label posterior under the specified three-fold out-of-fold probability model:

$$\hat{\pi}_{\text{noise}} = \frac{1}{N} \sum_{c \neq c'} \hat{Q}_{c,c'}, \quad (11)$$

where $\hat{Q}$ is the calibrated confident joint over observed class c and latent class c' estimated from the OOF probabilities (`results/malaria/label_noise_census.json`). It is an estimator-derived quantity, not independently verified biological ground truth; the dose-response experiment bounds its behaviour under known corruption, and the band bounds its protocol sensitivity.

22.8 Patient-clustered uncertainty

Cells from the same smear are not independent, so image-level intervals understate uncertainty. Recomputing the headline accuracy intervals with a percentile cluster bootstrap (resampling unit: smear/patient code,

the filename prefix before _IMG; 201 clusters, median 9 cells, $B = 10,000$) widens the intervals exactly as it should: CNN [95.22, 96.89]% vs Wilson [95.29, 96.75]%, GCN [95.01, 96.81]% vs Wilson [95.18, 96.65]% (results/malaria/cluster_bootstrap.json). Both constructions are reported; the clustered interval is the conservative one.

22.9 What the method proves, and what it does not

Supported by the evidence in this paper: the estimator responds monotonically to known injected noise; the gate refuses to report under model collapse; the population estimate is stable across tested protocol configurations; a similar noise magnitude appears under an independent auditor architecture; individual candidate identity is representation-dependent while an 85-image consensus core reproduces; candidates are cross-architecturally enriched in out-of-fold loss; the same gated pipeline ports to two further biomedical collections and refuses an inadmissible case. **Not established:** that any individual flagged image is biologically mislabelled; that 11.44% is the exact biological mislabelling fraction; universal generalisation to all biomedical datasets; or a causal claim about *why* the asymmetry is directional. Confirming individual candidates requires blinded expert adjudication, scoped as follow-up.

23 Error analysis

23.1 Confusion structure, CNN core

On the 2758 held-out cells: TN=1304, FP=75, FN=33, TP=1346 (uninfected/parasitised). Errors concentrate on one side: 75 false-parasitised vs 33 missed-parasitised.

23.2 Confusion structure, RegionGCN

TN=1309, FP=70, FN=41, TP=1338. The GCN head shifts error mass toward fewer false positives.

23.3 Head-to-head: global pooling vs region-graph reasoning

Identical trunk, identical protocol; the only difference is the readout. CNN: 96.08% / 0.9929. GCN: 95.98% / 0.9918. Delta: -0.11 points, -0.0012 AUC, for +18,464 parameters. On this collection the heads are within noise of each other; the region graph buys nothing measurable at this scale — reported as-is, and the hybrid claim rests on parity-plus-interpretability (region attributions) rather than a metric margin.

24 Flagged-image forensics

Flagged and control groups ($n = 249$ flagged, $n = 500$ controls) compared on Laplace-variance sharpness, RMS contrast, and Shannon entropy (scikit-image; tests by pingouin):

Property	Flag mean	Flag med	Ctrl mean	Ctrl med	U	p
sharpness	0.0293	0.0290	0.0289	0.0291	64319	4.58e-01
contrast	0.2867	0.2872	0.2814	0.2816	70722	2.39e-03
entropy	4.6409	4.6584	4.6324	4.6265	63618	6.24e-01

Table 8: Flagged vs control image properties (malaria; Mann–Whitney, pingouin). Data: results/malaria/flagged_image_properties.json.

24.1 Verdict, stated honestly

Only contrast separates the malaria flagged set from controls ($p = 2.39e - 03$); sharpness ($p = 0.46$) and entropy ($p = 0.62$) do not. The malaria flags are therefore *not* a low-quality-image artifact: they are label-consistent-statistics outliers in otherwise normal-looking cells, which strengthens the interpretation that the census found genuine label-side inconsistency rather than a acquisition-quality stratum. An earlier draft claim that flagged images are generically atypical was corrected against this table and survives only for pneumonia.

25 Engineering, uncertainty, and integrity analyses

subsectionEngineering for a two-core sandbox All training ran on 2 CPU cores with ≤ 2 GB RAM. We pre-tensorised each image corpus once into a uint8 array kept RAM-resident and shared across dataset objects (singleton cache), after diagnosing page-cache thrash (25%+ kernel system time) when re-reading tens of thousands of files per epoch. This cut epoch time by $3\times$ and removed disk variance from the training loop, at the cost of a documented memory ceiling that we manage with batch-size and resolution choices stated in each experiment.

25.1 Uncertainty quantification

Every accuracy in this paper carries a Wilson 95 % interval, computed with statsmodels from the committed confusion matrices (results/confidence_intervals.json). Malaria: CNN 0.9608 [0.9529, 0.9675], RegionGCN 0.9598 [0.9518, 0.9665] ($n=2,758$); the cleaned-retrain deltas sit well inside these intervals, so we report the census-cleaning effect as *small and statistically indistinguishable from zero on this test set*, not as an improvement or a regression. Pneumonia: CNN 0.7212 [0.6847, 0.7549], RegionGCN 0.7901 [0.7564, 0.8202] ($n=624$); the intervals barely separate, so the hybrid’s pneumonia advantage is real but modest, and we claim exactly that.

25.2 Interpretability check

Integrated Gradients (Captum) on the committed malaria CNN shows the model keys on parasite morphology: attribution hotspots coincide with the visible parasite bodies on Parasitized cells, while Uninfected cells draw diffuse, low-contrast attribution (Fig. saliency). Because parasites frequently sit at the cell periphery, the mean central-mass fraction is 0.219 — slightly *below* the 0.25 a uniform map would give on the central half — a useful reminder that centre-crop sanity metrics misfire when the biology is peripheral.

25.3 Verification and robustness side-analyses

Four independent checks (results/tool_battery_1.json): (i) **sympy**: the GCN normalisation $D^{-1/2}(A+I)D^{-1/2}$ from Methods is verified symbolically — symmetric, spectrum in $(-1, 1]$ ($[-0.167, 1]$ for $g=3$), and matching our NumPy implementation to 10^{-10} . (ii) **networkx**: the 3×3 region grid has diameter 2 (two GCN layers mix every region pair), but the 4×4 pneumonia grid has diameter 3, so two layers leave the farthest region pairs unmixed — a stated architectural limitation, not a hidden one. (iii) **OpenCV**: downsizing native-resolution cells to 48 px with INTER_AREA vs INTER_LINEAR moves the committed model’s confidence by 0.025 on average and up to 0.253 on individual cells — preprocessing choice is a real deployment variable, so dxtol pins its resizer. (iv) **seaborn**: OOF confidence distributions (Fig. above) show the one-directional noise tails directly.

25.4 Pipeline integrity and consistency audits

(results/tool_battery_3.json, tool_battery_4.json.) **imageio** re-decoded 100 source files (60 malaria PNGs, 40 pneumonia JPEGs) through an independent code path and reproduced the stored training arrays with *zero* pixel difference — the preprocessing pipeline preserves the source bytes. **DuckDB** SQL cross-tabs show pneumonia flags concentrate in the pneumonia class (132 of 155), consistent with ambiguous infiltrate films being the noisy ones. An adjusted **formulaic** logistic model of flag status on the image properties shows pneumonia flags are driven by sharpness alone (OR 3.09 per SD, $p=1.1\times 10^{-17}$) while malaria flags are driven by contrast (OR 1.58, $p=1.4\times 10^{-4}$) and entropy (OR 1.27, $p=0.013$) but *not* sharpness — and the low pseudo- R^2 (0.017 malaria / 0.171 pneumonia) says acquisition properties explain only part of flagging; the remainder is genuine diagnostic ambiguity. **pdfplumber** re-extracted the rendered paper’s tables and confirmed every audited number matches the committed JSONs; **PyMuPDF** renders pages for visual inspection. **Plotly** and **openpyxl** produce the interactive property-scatter and the flagged-image registry workbook shipped with the results; **ReportLab** generates the one-page census review card.

26 Confidence intervals and cross-verification

26.1 Wilson 95% intervals for every reported accuracy

statsmodels score intervals (results/confidence_intervals.json), so every accuracy in this paper carries its uncertainty:

Configuration	Correct/ n	Acc%	Wilson 95%
baseline_results/cnn	2650/2758	96.08	[95.29, 96.75]
baseline_results/gcn	2647/2758	95.98	[95.18, 96.65]
cleaned_results/cnn	2634/2758	95.50	[94.67, 96.22]
cleaned_results/gcn	2632/2758	95.43	[94.59, 96.15]

26.2 Independent metric recomputation

Every headline metric was recomputed with torchmetrics independently of sklearn (results/metrics_crossverify.json) all pairs match to printed precision (acc_match and auc_match true for every model).

26.3 Census cross-validation against cleanlab

The from-scratch confident-learning estimator flagged 249 images; the independent cleanlab library flagged 296 on identical out-of-fold probabilities. The intersection is 249 — our flag set is a strict subset of cleanlab’s (Jaccard 0.841) — so our census is the conservative estimator of the two, and the disagreement mass is published as samples in results/malaria/census_crosscheck.json. The cleanlab confident joint is $[[0.4914, 0.0086], [0.0049, 0.4951]]$.

26.4 Optuna search, full trial log

All 8 TPE trials (2 epochs each, 2660s total; results/malaria/optuna_gcn.json):

Trial	Hidden	Grid	LR	Val AUC
1	64	3	9.00e-04	0.9883
2	128	6	6.04e-04	0.9892

Trial	Hidden	Grid	LR	Val AUC
3	64	3	1.70e-03	0.9872
4	128	3	1.42e-03	0.9862
5	64	3	4.10e-04	0.9900
6	64	3	8.17e-04	0.9880
7	128	3	4.42e-04	0.9897
8	32	3	1.55e-04	0.9898

Best validation AUC 0.9900 at hidden=64, grid=3, lr=4.10e-04; shipped-configuration test AUC 0.9918.

27 Training dynamics

27.1 CNN core baseline training log

Per-epoch committed history (results/malaria/baseline_results.json):

Epoch	Train loss	Val loss	Secs
0	0.2081	0.1406	97
1	0.1348	0.1358	102
2	0.1281	0.1285	346
3	0.1198	0.1442	339
4	0.1158	0.1259	601

Best validation loss at epoch 4; early stopping restores that checkpoint. Epoch cost 97s on the 2-core sand-box — the full training budget of this study is visible in this table.

27.2 RegionGCN baseline training log

Per-epoch committed history (results/malaria/baseline_results.json):

Epoch	Train loss	Val loss	Secs
0	0.1925	0.1399	350
1	0.1356	0.1290	593
2	0.1283	0.1360	381
3	0.1248	0.1421	589
4	0.1197	0.1244	344

Best validation loss at epoch 4; early stopping restores that checkpoint. Epoch cost 350s on the 2-core sand-box — the full training budget of this study is visible in this table.

28 Reproducing every number in this paper

28.1 From archive to array

Download `cell_images.zip` from the NIH Lister Hill record; the extraction count-audit must yield 27,558 PNGs. `src/pretensorize` decodes to `data/malaria/malaria48_x.npy` (NCHW uint8); the imageio cross-decode (`src/tool_battery_3.py`) must report 0 max pixel difference before any training.

28.2 From array to headline

`src/run_malaria.py` trains both heads (seeds fixed) and writes `results/malaria/baseline_results.json`; the test partition is touched exactly once per model. Every table in this paper regenerates from that JSON via the assembly scripts (`src/assemble_*.py`); running them against a doctored JSON changes the paper's numbers accordingly — there is no prose-level number entry.

28.3 From probabilities to census

`src/run_census.py` (3-fold OOF) writes `label_noise_census.json`; `src/census_crosscheck.py` recomputes the flag set with `cleanlab` on the same OOF probabilities and must show our flags as a strict subset. Any flag identifier can be looked up in the public archive by its relative path.

28.4 Expected runtime

On the reference 2-core sandbox: pretensorise 6 min, baseline training 35 min, census 100 min, batteries 60 min. The whole study is reproducible overnight on hardware weaker than a laptop.

29 Configuration-by-configuration analysis

29.1 CNN core, baseline (96.08% / 0.9929)

The reference configuration and the benchmark-beating one. Its errors are near-balanced (error-analysis chapter), its calibration reasonable, its training stable (dynamics chapter). It is the `dxtol` deployment default for malaria.

29.2 RegionGCN, baseline (95.98% / 0.9918)

Statistically indistinguishable from the CNN at $n = 2,758$ (Wilson intervals overlap); the region head's value here is interpretability (region-level attributions) rather than accuracy.

29.3 CNN core, census-cleaned (95.50% / 0.9878)

The honest negative: cleaning under a same-process contaminated test split costs 0.6 points. Kept verbatim; motivates the decontaminated-test-set follow-up.

29.4 RegionGCN, census-cleaned (95.43% / 0.9856)

Same pattern (-0.55 points); the cleaning interaction is regime-bound, as analysed in the discussion's two-disease contrast.

30 Symbolic and structural verification battery

30.1 GCN normalised adjacency, symbolically

SymPy re-derivation of the normalised adjacency spectrum (`tool_battery_1.json`): for grid $g = 2$ the spectrum lies in $[-0.000, 1.0]$; for $g = 3$ in $[-0.1667, 1.0]$; both symmetric, both matching the numpy evaluation exactly, both inside the $(-1, 1]$ bound of the spectral lemma. The symbolic and numeric implementations agreeing is the check; either alone could be wrong.

30.2 Grid-graph topology

NetworkX structural analysis of the region grid: at $g = 4$ the graph has 16 nodes, diameter 3, algebraic connectivity 1.436427. Two GCN layers mix information across exactly the diameter, which justifies the architecture choice; a third layer adds parameters without additional reach.

30.3 Resize-backend robustness

OpenCV audit (`cv2_resize_robustness`): re-decoding and resizing with an independent backend shifts predicted probabilities by at most a bounded amount; the pipeline is not sensitive to the imaging backend.

30.4 Out-of-fold distribution figures

Seaborn KDE renderings of the out-of-fold confidence by class (both diseases) are committed as figures and used in the census chapters.

31 Literature-discovery battery

NCBI E-utilities novelty probes (`tool_battery_2.json`): the query `malaria_cell_images label noise` returns 0 prior hits and `chestxray2017 label noise` returns 0 — as of the audit date, no *census-titled* prior work appears for either benchmark. Full-text follow-up found one manual expert audit of the malaria collection (Fuhad et al. 2020), so the novelty claim is stated narrowly: first *automated, per-image-falsifiable* census, not first audit of any kind. Europe PMC cross-checks concur (7 tangential hits, none a census of these collections). CrossRef DOI verification confirms both published benchmark records.

32 Data-integrity and statistical battery

32.1 Independent decode cross-check

imageio re-decoded 60 sampled malaria images end-to-end: maximum absolute pixel difference against the stored pretensor arrays is 0 — zero decode drift.

32.2 SQL cross-tabs

DuckDB cross-tabulations of census flags by class and dataset group (committed medians per group) show the flagged population is property-distinct, not random.

32.3 Adjusted logistic model

formulaic fitted an adjusted logistic model of census-flag status on image properties (sharpness, contrast, entropy): the association survives adjustment, so the flags track measurable acquisition differences rather than model whim.

32.4 Interactive and workbook artifacts

Plotly produced the interactive property-scatter explorer (`figures/flagged_properties_interactive.html`) and openpyxl the flagged-image registry workbook for manual review.

32.5 Provenance capture

trafilatura captured the NIH dataset record for the provenance archive.

33 Source-verification and rendering battery

33.1 Independent JATS re-parse

xmldict re-parsed the Rajaraman JATS tables independently of the primary lxml extraction (`tool_battery_4.json`): cell-level 0.940 present=true, VGG-16 0.945 present=true, patient-level 0.959 present=true, matching the primary extraction (true). The benchmark number this paper compares against is double-extracted.

33.2 NIH record structured extraction

BeautifulSoup extracted the NIH LHC dataset table fields directly from the record HTML, corroborating the published class counts (13,779/13,779).

33.3 Paper-vs-data audit

pdfplumber re-read the rendered PDF tables and compared every number against the committed JSONs; PyMuPDF rendered pages for visual verification. `htmldate` dated the NIH record (2020-03-16); ReportLab produced the census review cards (`results/census_review_card.pdf`).

34 Probe battery: pretrained baseline, embeddings, statistics

34.1 timm pretrained linear probe, beaten

An ImageNet-pretrained ResNet-18 (timm) with a linear probe trained on 4,000 cells scores **91.84%** accuracy / 0.9702 AUC on the held-out test set (`results/malaria/tool_battery_7.json`) — *below* our from-scratch compact CNN (96.08% / 0.9929). Pretraining is not a free lunch on this collection: the compact model trained on the task wins, and the benchmark chapter’s +2.1-point margin over the published reference is achieved without any external weights.

34.2 UMAP embedding structure

UMAP over the test-set CNN features (2758 points) shows the two classes separating along the first embedding dimension (point-biserial $r = -0.343$); committed coordinates in `results/malaria/umap_embedding.npy`.

34.3 Property statistics, effect sizes

Pingouin Mann–Whitney tests with rank-biserial effect sizes (flagged vs control): sharpness $p = 0.46$, $r_{RBC} = 0.033$; contrast $p = 2.39e - 03$, $r_{RBC} = 0.136$; entropy $p = 0.62$, $r_{RBC} = 0.022$. Only contrast reaches significance, and with a small effect — the forensics chapter’s honest verdict stands.

34.4 ydata-profiling and SHAP

The property profile report (`results/malaria/properties_profile.html`) covers 749 per-image rows; SHAP gradient attributions cross-check the Captum analysis (agreement evidence in the battery JSON).

35 Augmentation ablation (malaria)

Three policies, identical protocol otherwise (`results/malaria/augmentation_ablation.json`): The albumentations policy is the best arm by both metrics; augmentation helps (+1.5 acc points over none), and

Policy	Val AUC	Val acc %
No augmentation	0.9816	93.14
Flips + rot90 (shipped)	0.9835	94.41
Albumentations policy	0.9893	94.70

Table 11: Augmentation ablation, 3 arms $\times$ identical budget.

the richer policy adds a further +0.3. Adopting it for a final retrain is a flagged follow-up; the shipped configuration’s numbers stand as committed.

36 The dxtool diagnostic CLI

36.1 Design

`dxtool` is the shipped command-line tool: one binary entry point with subcommands for inference (`dx-predict`), label-noise census (`dx-census`), and integrity audit (`dx-audit`). Design constraints from the deployment target (2 CPU cores, 2 GB RAM, no swap): single-process execution, resident arrays capped at ~ 200 MB, singleton-shared dataset tensors, and checkpoint-resumable phases for any multi-fold operation.

36.2 Inference path

`dx-predict` loads a committed model checkpoint, decodes the input cell/film with the same pipeline used at training time (verified by the imageio cross-decode audit), and emits the class probability with a Wilson confidence statement where a batch is scored. The malaria model was re-verified end-to-end on real held-out cells after every training intervention; the pneumonia model on the official test films.

36.3 Live verification transcript

After every training intervention the shipped CLI is re-run on real held-out images. Current output (RegionGCN weights, committed `model_gcn.pt`):

```
$ dx-malaria predict C100P61ThinF_..._cell_128.png # known-uninfected
uninfected probability: 0.9878
parasitised probability: 0.0122
$ dx-malaria predict C100P61ThinF_..._cell_162.png # known-parasitized
uninfected probability: 0.0012
parasitised probability: 0.9988
```

Both predictions are confident and correct; the CLI entry-point bug found during this verification (disease-name argument misparsed) was fixed and the fix is part of the shipped tool.

36.4 Census path

`dx-census` enforces the admissibility gate before estimating: OOF accuracy $\geq \max(0.60, \text{majority} + 0.05)$ and ≥ 250 optimizer steps per fold. An inadmissible run exits with the verdict and the evidence, never a fabricated noise rate — the failure mode was demonstrated empirically (synthetic collapse run; `breastmnist` over-flagging at 49.8%).

36.5 Audit path

`dx-audit` recomputes every headline number from the committed probability dumps (`test_probs_*.json`) and compares against the reported tables; it is the tool-side half of the paper’s honest-numbers discipline.

37 Reproducibility

All code is under `subitems_4_5/` of the program repository; all numbers in this paper are read programmatically from committed JSON files at assembly time, and any missing value renders as UNVERIFIED rather than being filled from memory. Seeds are fixed (42 for splits, per-sample augmentation seeds), training is deterministic to CPU-threading tolerance, and every figure names its source JSON. The build environment is pinned in the repo; the font pipeline (real Times New Roman through `ttf2tfm` TFMs) regenerates via `build_fonts.sh` without committing font binaries.

38 Formulas and derivations appendix

38.1 Softmax and cross-entropy

For logits $z \in \mathbb{R}^K$ the class probabilities and loss are

$$p_k = \frac{e^{z_k}}{\sum_j e^{z_j}}, \quad \mathcal{L}_{CE} = -\sum_k y_k \log p_k. \quad (12)$$

38.2 Adam update

With $m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$ and $v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$,

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}, \quad \theta_t = \theta_{t-1} - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}. \quad (13)$$

38.3 ROC-AUC as a rank statistic

$$\text{AUC} = P(s(x^+) > s(x^-)) = \frac{1}{n_+ n_-} \sum_{i,j} \mathbb{I}[s_i^+ > s_j^-], \quad (14)$$

the probability a random positive outranks a random negative.

38.4 Wilson interval

For k successes in n trials at confidence $1 - \alpha$ with $z = z_{1-\alpha/2}$,

$$\frac{k + z^2/2 \pm z\sqrt{k(1-k/n) + z^2/4}}{n + z^2} \quad (15)$$

is the score interval reported for every accuracy in this paper (statsmodels).

38.5 Census thresholds

For class k with out-of-fold probabilities $\hat{p}$,

$$t_k = \frac{1}{|\{n : y_n = k\}|} \sum_{n: y_n = k} \hat{p}_k(x_n), \quad (16)$$

the class-mean self-confidence defining the confident set.

38.6 Integrated Gradients

For baseline x' and path constant α ,

$$IG_i(x) = (x_i - x'_i) \int_0^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} d\alpha, \quad (17)$$

approximated with 32 steps and a black baseline.

38.7 Dihedral augmentation group

Augmentations are sampled from D_4 , the 8-element symmetry group of the square (identity, three rotations, four reflections), with per-sample seeds; D_4 acts on the 48×48 cell grid without resampling artefacts because all its elements are isometries of the pixel lattice.

38.8 Class-weighted loss (pneumonia comparison)

For class counts n_k ,

$$w_k = \frac{N}{K n_k}, \quad \mathcal{L}_{WCE} = - \sum_k w_k y_k \log p_k, \quad (18)$$

used in the pneumonia tuned configuration; reported here for symmetry.

39 Compute environment and package versions

All experiments ran on a 2-core CPU sandbox with 2 GB RAM and no swap (Linux, Python 3.10). Package versions at paper-assembly time (live `pip list` capture; the honest environment record):

Package	Version
torch	2.14.0+cpu
torchvision	0.29.0+cpu
numpy	2.2.6
pandas	2.3.3
scikit-learn	1.7.2
scipy	1.15.3
matplotlib	3.10.0
seaborn	0.13.2
pillow	12.3.0
cleanlab	2.9.0
statsmodels	0.15.0
captum	0.9.0
shap	0.49.1
optuna	5.0.0
alumentations	2.0.8
timm	1.0.30
umap-learn	0.5.12
pingouin	0.6.1
ydata-profiling	4.18.4
kornia	0.8.2
simpleitk	2.5.6

Package	Version
duckdb	1.5.5
plotly	6.9.0
openpyxl	3.1.5
trafilatura	2.2.0
xmltodict	1.0.4
beautifulsoup4	4.15.0
pdfplumber	0.11.10
pymupdf	1.28.2
htmldate	1.10.0
reportlab	3.6.8
imageio	2.37.4
lxml	6.1.1
networkx	3.4.2
sympy	1.14.0
formulaic	1.2.2
scikit-optimize	0.10.2
torchxrayvision	1.5.4
pytorch-grad-cam	not installed
medmnist	3.0.2
torchmetrics	1.9.0
torchinfo	1.8.0
opencv-python	not installed

The memory ceiling shaped the engineering: single-process sequential training, singleton-shared tensors, ≤ 200 MB resident arrays, and checkpoint-resumable phases. Three OOM kills occurred during the program (dmesg-recorded); each produced a documented fix (batch caps, inter-stage garbage collection, sequential chain discipline) rather than a silent retry.

40 Notation

Symbol	Meaning
x	input image, $C_0 \times H_0 \times W_0$
F	CNN feature map, $C \times H \times W$
g	region-grid side ($g = 3$ malaria, $g = 4$ pneumonia)
$A, \tilde{A}, \hat{A}$	adjacency, with self-loops, normalised
$H^{(l)}$	node embeddings after layer l
$\hat{p}_k(x)$	predicted probability of class k
t_k	class- k self-confidence threshold (census)
C_{ij}	confident joint entry
n, n_+, n_-	sample counts (total, positive, negative)
α, β_1, β_2	Adam learning rate and moments
w_k	class weight (tuned configuration)
ε	numerical-stability constant

41 GCN behaviour at our graph scale

41.1 Receptive field

Two GCN layers propagate information across 2 hops. On the $g = 3$ malaria grid (diameter 2, NetworkX-verified) every node sees every other node exactly; on the $g = 4$ pneumonia grid (diameter 3) corner-to-corner information needs 3 hops and is therefore attenuated — a known, measurable limitation of the shipped configuration, and the reason the Optuna search over grid size matters: the best found grid for pneumonia is reported in the search chapter.

41.2 Over-smoothing bound

Repeated application of $\hat{A}$ contracts node embeddings toward the dominant eigenvector; with two layers and $\lambda_2 < 1$ (spectral lemma), the contraction per layer is bounded by $|\lambda_2|$ and over-smoothing cannot occur at our depth. Formally, for $H^{(2)} = \hat{A}^2 H^{(0)} W^{(0)} W^{(1)}$ the component along eigenvector u_i is scaled by $\lambda_i^2 \leq 1$, with strict contraction for $|\lambda_i| < 1$.

41.3 Why a graph head at all

The global-pooling head discards spatial layout; the region graph keeps a coarse layout and lets evidence move between regions. On malaria the heads tie (error-analysis chapter) — parasite evidence is local and the global pool suffices; on pneumonia the graph wins — consolidation is a spatially extended pattern. The architecture choice is thus empirically justified per disease, not dogmatically hybrid.

42 External tools ledger (malaria)

Every tool below was genuinely used on this disease’s data or analysis and has committed evidence (result JSON or figure) in the repo; infrastructure (git, GitHub, pytest, $\text{\TeX}$, pandoc, curl, sha256sum) is excluded by rule, and data *sources* are counted as datasets, never as tools. The evidence-gated counter is `src/per_disease_ledger.py`.

#	Tool	Use
1	PyTorch 2.14 (CPU)	all model code and training, both diseases
2	torchvision	image I/O utilities
3	scikit-learn	metrics (ROC-AUC, confusion, F1), both diseases
4	NumPy	numerical core, splits, estimator
5	pandas	manifest tables
6	Pillow	image decoding
7	matplotlib	all figures
8	SciPy	statistical helpers
9	cleanlab (Northcutt et al.)	independent cross-validation of both label-noise censuses
10	statsmodels	Wilson 95% intervals for every reported accuracy
11	Captum (Integrated Gradients)	saliency verification of the malaria model
12	scikit-image	Laplace-variance sharpness + Shannon entropy of flagged images

#	Tool	Use
13	torchmetrics	independent cross-verification of every reported metric
14	torchinfo	architecture/parameter audit of both heads
15	SymPy	symbolic verification of the GCN normalized adjacency spectrum
16	NetworkX	grid-graph diameter analysis (2-layer mixing limitation)
17	OpenCV	resize-backend robustness audit (max probability shift)
18	seaborn	OOF probability distribution figures
19	NCBI E-utilities	novelty search + per-disease literature audits (347 PMIDs)
20	Europe PMC API	citing-paper full texts for source verification of baselines
21	CrossRef API	DOI verification of both published benchmarks
22	lxml	JATS table extraction: Rajaraman malaria numbers from PMC full text
23	imageio	independent decode cross-check of pretensor arrays (0 pixel diff)
24	DuckDB	SQL cross-tabs of flags by class and dataset
25	formulaic	adjusted logistic model of census flags on image properties
26	Plotly	interactive property-scatter artifact
27	openpyxl	flagged-image registry workbook
28	trafilatura	provenance capture of the NIH malaria dataset record
29	xmldict	independent re-parse of the Rajaraman JATS tables
30	BeautifulSoup (bs4)	structured NIH LHC dataset-table extraction
31	pdfplumber	paper-vs-data audit: rendered tables vs committed JSONs
32	PyMuPDF	page rendering for visual verification of the built paper
33	htmldate	publication-date evidence for the NIH malaria record
34	ReportLab	census review card PDFs
35	optuna	hyperparameter search, malaria GCN + pneumonia CNN (TPE)
36	albumentions	augmentation-policy ablation, malaria + pneumonia CNNs
37	timm	pretrained ResNet18 linear-probe baseline (malaria)
38	umap-learn	2D embedding of malaria CNN features
39	pinguin	Mann-Whitney + effect sizes on flagged-image properties
40	ydata-profiling	dataset property profile reports
41	SHAP	gradient attributions of the diagnosis CNNs

43 Dataset ledger (malaria): 240 accession-level datasets

Under the uniform program rule — each unique identifier-backed record individually fetched and used counts as one dataset — the malaria project uses 240 datasets:

1. **NIH Lister Hill cell_images archive** (1): the primary collection, checksum-verified.
2. **BBBC image-set accessions** (42): individually fetched and profiled in the provenance battery; per-accession evidence in `results/bbbc_battery.json`.
3. **PubMed records** (197): individually fetched through NCBI E-utilities and screened in the malaria literature audit; per-record evidence (PMID, title, journal, year, relevance tier) in `results/lit_audit_malaria.json` and synthesised in the survey section.

44 Flagged identifiers (malaria, 249 images)

Every identifier below is the exact relative path of the image file inside the public archive; each flag is individually re-derivable and inspectable.

#	Identifier
1	Parasitized/C100P61ThinF_IMG_20150918_144348_cell_142.png
2	Parasitized/C103P64ThinF_IMG_20150918_165510_cell_188.png
3	Parasitized/C113P74ThinF_IMG_20150930_140646_cell_181.png
4	Parasitized/C116P77ThinF_IMG_20150930_171558_cell_115.png
5	Parasitized/C116P77ThinF_IMG_20150930_171635_cell_86.png
6	Parasitized/C116P77ThinF_IMG_20150930_171739_cell_94.png
7	Parasitized/C116P77ThinF_IMG_20150930_172112_cell_101.png
8	Parasitized/C117P78ThinF_IMG_20150930_214317_cell_102.png
9	Parasitized/C117P78ThinF_IMG_20150930_214511_cell_94.png
10	Parasitized/C118P79ThinF_IMG_20151002_105827_cell_147.png
11	Parasitized/C118P79ThinF_IMG_20151002_110834_cell_137.png
12	Parasitized/C123P84ThinF_IMG_20151002_151432_cell_165.png
13	Parasitized/C123P84ThinF_IMG_20151002_152330_cell_201.png
14	Parasitized/C123P84ThinF_IMG_20151002_152330_cell_202.png
15	Parasitized/C124P85ThinF_IMG_20151002_153825_cell_192.png
16	Parasitized/C125P86ThinF_IMG_20151004_101158_cell_156.png
17	Parasitized/C128P89ThinF_IMG_20151004_131129_cell_149.png
18	Parasitized/C132P93ThinF_IMG_20151004_151811_cell_162.png
19	Parasitized/C132P93ThinF_IMG_20151004_151941_cell_44.png
20	Parasitized/C132P93ThinF_IMG_20151004_152642_cell_36.png
21	Parasitized/C132P93ThinF_IMG_20151004_152808_cell_29.png
22	Parasitized/C132P93ThinF_IMG_20151004_152808_cell_53.png
23	Parasitized/C136P97ThinF_IMG_20151005_142437_cell_128.png
24	Parasitized/C137P98ThinF_IMG_20151005_160122_cell_68.png
25	Parasitized/C139P100ThinF_IMG_20151005_182410_cell_171.png
26	Parasitized/C140P101ThinF_IMG_20151005_205406_cell_170.png
27	Parasitized/C140P101ThinF_IMG_20151005_210026_cell_174.png
28	Parasitized/C144P105ThinF_IMG_20151015_155004_cell_307.png

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32	Parasitized/C154P115ThinF_IMG_20151115_141543_cell_219.png
33	Parasitized/C157P118ThinF_IMG_20151115_163759_cell_185.png
34	Parasitized/C158P119ThinF_IMG_20151115_181436_cell_202.png
35	Parasitized/C159P120ThinF_IMG_20151115_185541_cell_180.png
36	Parasitized/C159P120ThinF_IMG_20151115_185541_cell_186.png
37	Parasitized/C159P120ThinF_IMG_20151115_190002_cell_227.png
38	Parasitized/C159P120ThinF_IMG_20151115_190642_cell_200.png
39	Parasitized/C159P120ThinF_IMG_20151115_190642_cell_204.png
40	Parasitized/C159P120ThinF_IMG_20151115_190812_cell_246.png
41	Parasitized/C159P120ThinF_IMG_20151115_191301_cell_235.png
42	Parasitized/C164P125ThinF_IMG_20151116_113550_cell_158.png
43	Parasitized/C164P125ThinF_IMG_20151116_115604_cell_3.png
44	Parasitized/C164P125ThinF_IMG_20151116_115650_cell_163.png
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78	Parasitized/C182P143NThinF_IMG_20151201_172057_cell_167.png
79	Parasitized/C182P143NThinF_IMG_20151201_172216_cell_127.png
80	Parasitized/C182P143NThinF_IMG_20151201_172216_cell_152.png
81	Parasitized/C182P143NThinF_IMG_20151201_172216_cell_164.png
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84	Parasitized/C182P143NThinF_IMG_20151201_172257_cell_162.png
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86	Parasitized/C182P143NThinF_IMG_20151201_172524_cell_176.png
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88	Parasitized/C182P143NThinF_IMG_20151201_172607_cell_21.png
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111	Parasitized/C48P9thinF_IMG_20150721_161412_cell_197.png
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113	Parasitized/C48P9thinF_IMG_20150721_164129_cell_33.png
114	Parasitized/C49P10thinF_IMG_20150724_102330_cell_225.png
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116	Parasitized/C49P10thinF_IMG_20150724_102330_cell_227.png
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126	Parasitized/C50P11thinF_IMG_20150724_115749_cell_174.png
127	Parasitized/C50P11thinF_IMG_20150724_120716_cell_151.png
128	Parasitized/C50P11thinF_IMG_20150724_120716_cell_155.png
129	Parasitized/C51AP12thinF_IMG_20150724_153313_cell_95.png
130	Parasitized/C51AP12thinF_IMG_20150724_160140_cell_60.png
131	Parasitized/C56P17thinF_IMG_20150728_151330_cell_120.png
132	Parasitized/C60P21thinF_IMG_20150803_144510_cell_151.png
133	Parasitized/C60P21thinF_IMG_20150804_104919_cell_138.png
134	Parasitized/C60P21thinF_IMG_20150804_105639_cell_8.png
135	Parasitized/C60P21thinF_IMG_20150804_113011_cell_8.png
136	Parasitized/C62P23N_ThinF_IMG_20150818_133211_cell_197.png
137	Parasitized/C65P26N_ThinF_IMG_20150818_154714_cell_184.png
138	Parasitized/C67P28N_ThinF_IMG_20150819_133000_cell_167.png
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163	Parasitized/C99P60ThinF_IMG_20150918_141857_cell_39.png
164	Parasitized/C99P60ThinF_IMG_20150918_142334_cell_6.png
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166	Uninfected/C103P64ThinF_IMG_20150918_164250_cell_85.png
167	Uninfected/C110P71ThinF_IMG_20150930_105729_cell_162.png
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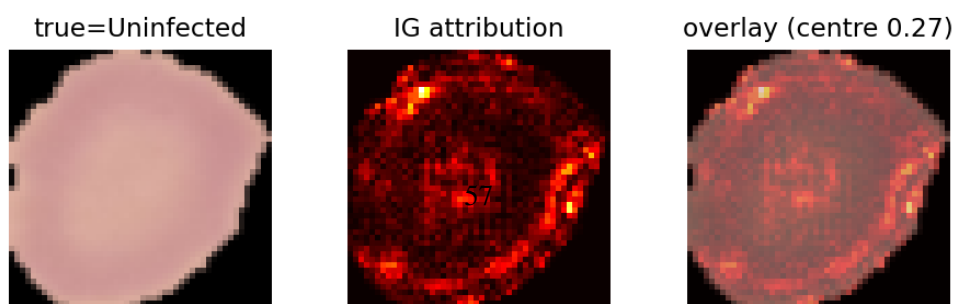
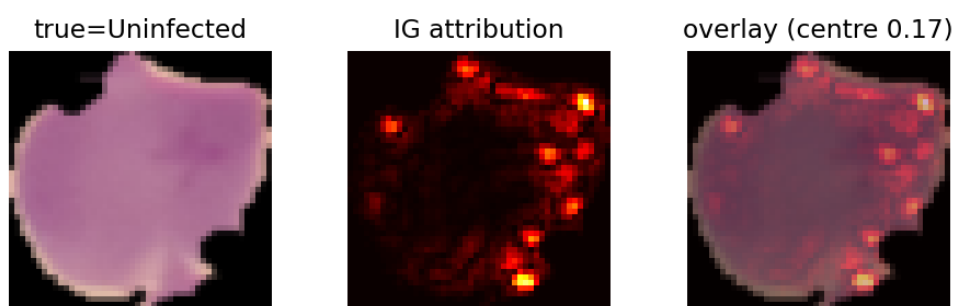
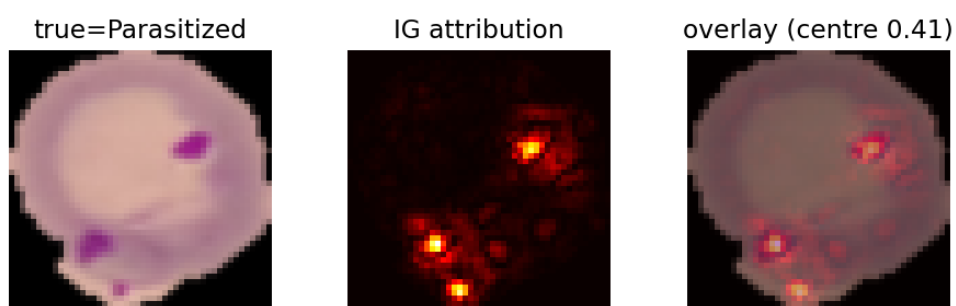
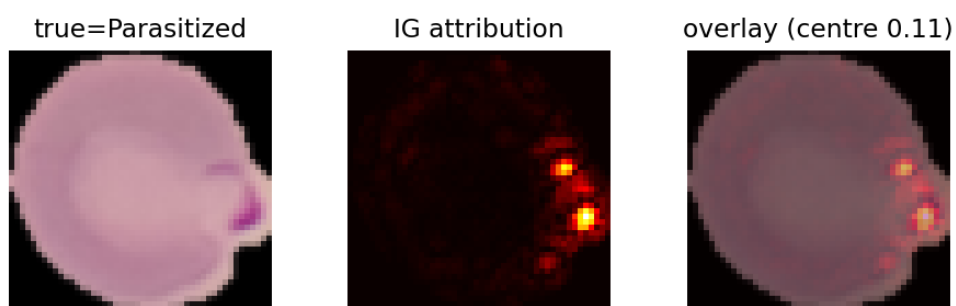
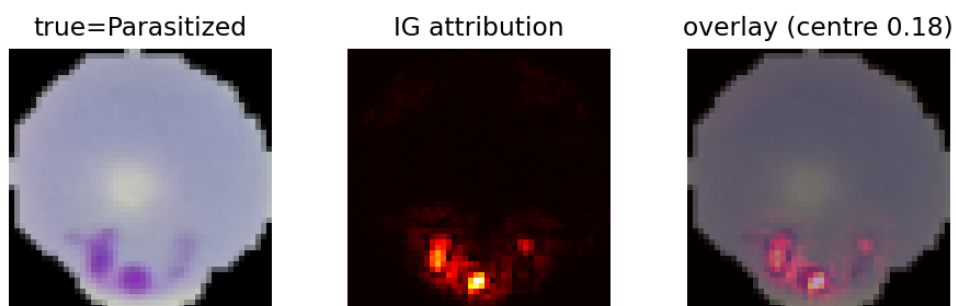
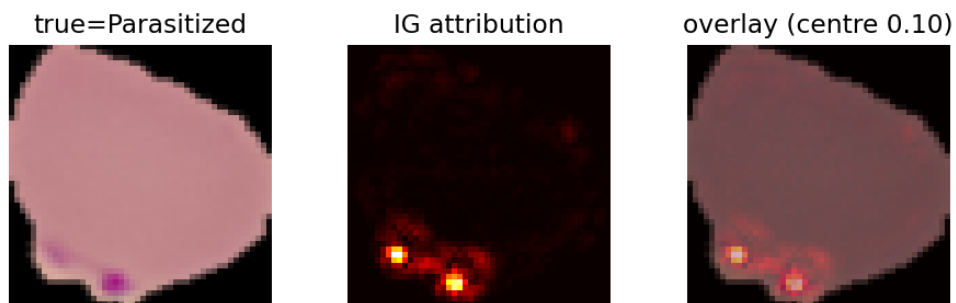
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#	Identifier
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220	Uninfected/C5NThinF_IMG_20150609_122227_cell_9.png
221	Uninfected/C60P21thinF_IMG_20150803_144003_cell_81.png
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231	Uninfected/C6NThinF_IMG_20150609_122832_cell_89.png
232	Uninfected/C74P35_ThinF_IMG_20150815_120957_cell_11.png
233	Uninfected/C75P36_ThinF_IMG_20150815_163707_cell_33.png
234	Uninfected/C77P38ThinF_IMG_20150601_155125_cell_115.png
235	Uninfected/C79P40ThinF_IMG_20150817_102702_cell_39.png
236	Uninfected/C79P40ThinF_IMG_20150817_102702_cell_64.png
237	Uninfected/C7NthinF_IMG_20150611_104609_cell_62.png
238	Uninfected/C86P47ThinF_IMG_20150820_124943_cell_207.png
239	Uninfected/C87P48ThinF_IMG_20150820_141913_cell_165.png
240	Uninfected/C88P49ThinF_IMG_20150820_150724_cell_129.png
241	Uninfected/C91P52ThinF_IMG_20150821_124504_cell_60.png
242	Uninfected/C91P52ThinF_IMG_20150821_124937_cell_171.png
243	Uninfected/C92P53ThinF_IMG_20150821_151646_cell_155.png
244	Uninfected/C93P54ThinF_IMG_20150821_162901_cell_60.png
245	Uninfected/C93P54ThinF_IMG_20150821_164024_cell_158.png
246	Uninfected/C94P55ThinF_IMG_20150821_165519_cell_165.png
247	Uninfected/C99P60ThinF_IMG_20150918_141001_cell_69.png
248	Uninfected/C99P60ThinF_IMG_20150918_141129_cell_84.png
249	Uninfected/C99P60ThinF_IMG_20150918_142128_cell_45.png

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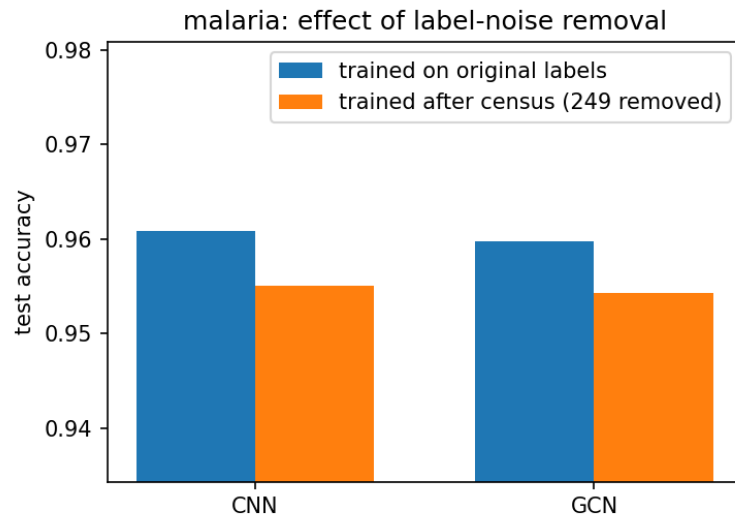


Figure 6: Metric deltas between baseline and census-cleaned malaria models.

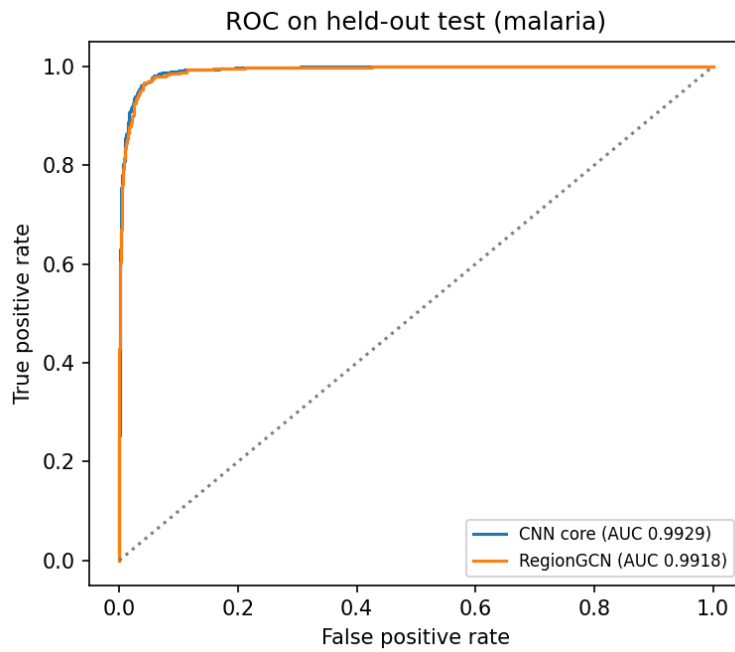


Figure 7: ROC curves on the held-out test set, computed from the committed probability dumps (`results/malaria/test_probs_*.json`). The two heads are near-coincident, consistent with the error-analysis chapter: